

# What's Driving High Electric Vehicle Fast Charging Prices?

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## Abstract

Electric vehicle fast charging is more costly than home charging, which could hinder governments from meeting electric vehicle adoption targets and place higher costs on lower income households that cannot charge at home. I compile novel data on station quantities, prices, and marginal operating costs to shed light on price setting behavior across charging networks. Using a difference-in-differences approach, I show that contrary to standard theory, stations do not lower prices in response to increased competition. I estimate a model of heterogeneous consumer demand for fast charging and use the elasticities from the demand model to calculate the separate sets of social surplus maximizing prices. I show that charging prices include a large markup above what is necessary to recover fixed costs, and lowering prices to the efficient prices would increase social surplus by 7-10%. Contrary to most settings, the optimal linear and two-part tariff prices that recover fixed costs are nearly the same. In the presence of charging station capacity constraints, equating quantity demanded and supplied requires increasing efficient prices above the marginal operating costs. These results suggest that increasing competition can only reduce markups by half, and policy would need to address charging supply or inelastic consumer demand to further reduce markups.

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# 1 Introduction

Many of the worlds’ largest car markets have set ambitious goals for electric vehicle (EV) adoption.<sup>1</sup> A hurdle in meeting these goals will be making EVs appealing for drivers who are unable to charge vehicles at home. In California, which is home to one third of United States EVs and the focus of this paper, nearly 90% of current EV drivers can charge at home, but only half of all drivers would be able to (Hoang and Davis, 2026). Many governments have focused on public fast chargers as the solution to this challenge. Fast chargers can refuel an EV in 20-60 minutes (compared to 4-50 hours at other chargers), allowing drivers without access to a charger at home or work to refuel. In 2024, California estimated it would need 83,000 fast chargers to support a fully electric car fleet – a notable increase from the 10,000 it had in 2023 – and has earmarked around \$350 million for fast charger subsidies since 2024 (Davis et al., 2024).

The premium speed comes at a premium price. Fast charging is on average three times as expensive as home charging in the United Sates (Paris, 2025). High charging prices have the potential to negatively impact both efficiency and equity. Prices above the social marginal cost could lead to deadweight loss from inefficiently low adoption and EV use for drivers without access to chargers at their home or workplace. The ability to charge at home is largely determined by whether a household has its own garage, which is highly correlated with income. In California, under a quarter of drivers who live in multi-family homes in disadvantaged or low-income communities could charge at home (Hoang and Davis, 2026).

In this paper, I explore the causes of high prices and estimate the impacts on social surplus. One barrier to research on pricing in this market is that networks typically share prices only through their own mobile applications, leading to an absence of centralized pricing data. I construct a novel weekly panel of hourly prices at the three largest networks in California – Tesla, EVgo, and Electrify America – by manually collecting data from networks’ mobile applications. The panel includes 19 months of week by hour price data for 58% of all charging stations in California. I build a second panel of EV charging volumes in Northern California using four years of electricity consumption data from Pacific Gas & Electric Company, one of the largest utilities in the United States.

I show that fast charging prices reflect large markups over marginal operating costs, which are an order of magnitude greater than those in the retail gasoline industry. I assess whether markups decrease as stations enter the market. I find that when a station enters, charging volumes and prices at nearby competitors remain unchanged. However, entry does

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<sup>1</sup>Although the United States has not set any targets, a number have their own and 13 states have formed the Affordable Clean Cars Coalition to help meet their targets.

decrease charging volumes by 14% at nearby stations in the same network, suggesting that consumers are willing to substitute within but not across networks. I show that a new Tesla station’s opening leads to a large reduction in the amount of time nearby Tesla stations are at capacity and a 3% decrease in prices at those stations.

Markups may provide benefits in the market. First, in the short run, networks cannot increase capacity and must rely on prices to achieve an efficient equilibrium. When the quantity demanded exceeds capacity, networks cannot guarantee that individuals with the highest willingness to pay are the ones that are served. This creates deadweight loss. Raising prices to equate quantity demanded and supplied eliminates the deadweight loss, so capacity constraints push efficient short-run prices above marginal operating costs. Second, networks may only build stations if they expect to earn profits, which requires recovering fixed costs. Marginal operating costs are flat, so networks will need to apply markups to recover fixed costs with linear prices. Economists generally prefer two-part tariff pricing for fixed cost recovery because firms can charge efficient prices and recover fixed costs through subscriptions. Each charging network encourages the use of subscriptions and has incentive to implement a two-part tariff pricing strategy to profit maximize.

I delve into network pricing strategies to determine how they set prices and how efficient observed prices are. I estimate a system of heterogeneous consumer demand for fast charging and use the own- and cross-price elasticities to solve the social surplus and profit maximization problems. I show that all three networks set prices near the linear prices that maximize charging market profit, suggesting they do not follow a two-part tariff strategy. This leads to a quantity-weighted average markup above marginal operating cost of 109%.

Markups are still present even at short-run efficient prices. I calculate that equating quantity demanded and supplied requires 48% markups above marginal operating costs and social surplus maximizing fixed-cost recovery with linear prices necessitates markups of 53-98%. I show that if observed prices were lowered to the prices that would most efficiently recover fixed costs, social surplus would increase by 7-10%. My results suggest policy encouraging price competition could create large benefits through lowering prices, though reduced profits could deter station entry. However, half of the observed markups are efficient given the current supply of chargers and consumer demand. My results suggest potentially large benefits from increasing the supply of charging or encouraging consumption at lower demand hours when stations are not capacity constrained.

I contribute to three literatures. First, I contribute to the literature on EV charging. A number of papers have estimated the effect of charger supply on EV purchases ([Remmy, forthcoming](#); [Cole et al., 2023](#); [Springel, 2021](#); [Sommer and Vance, 2021](#); [Zhou and Li, 2018](#); [Li et al., 2017](#)). My results are the first on the effect of charger supply on prices and charging

behavior. Importantly, I show that consumers exhibit strong network preferences that are not explained by charging standard. This is particularly relevant for work showing potentially large benefits of standardizing plug types (Li, 2023), which assumes potential drivers (rather than the current EV drivers I study) choose cars based on existing stations’ charging standard rather than network. My results suggest that current drivers choosing where to charge do not behave this way. I also provide the first estimates of demand elasticities for fast charging stations in the United States. This adds to randomized control trial evidence from Bernard et al. (2025) on elasticities of demand for public charging in the United Kingdom and Bailey et al. (2025) and Garg et al. (2026) on how drivers respond to price incentives for home and workplace charging, respectively.

Second, I contribute to the literature on market entry and competition. My results contrast the retail gasoline literature, where Davis, McRae and Seira (2025) and Taylor and Muehlegger (2025) have documented price decreases from incumbent stations in response to competitor entry. Similar to my results, Arcidiacono et al. (2020) find no effect of Walmart Supercenter entry on nearby grocery store prices. They find a negative impact on retail volumes and suggest that managerial inattention is the reason incumbents do not update prices. In contrast, EV drivers in California shift charging volumes from incumbents to new stations in the same network, but not different networks. These results suggests that consumer preferences or demand-side frictions, rather than the supply-side frictions noted by Arcidiacono et al. (2020), may explain the lack of price response in the fast charging market.

Third, I contribute to the literature on fixed-cost recovery and efficient pricing. “Ramsey prices” are the set of linear prices that minimize deadweight loss while recovering fixed costs (Boiteux, 1956; Ramsey, 1927). Ramsey prices still lead to deadweight loss, which means they are a second best option to two-part tariffs (Feldstein, 1972; Littlechild, 1975). I modify the standard Ramsey problem to account for short-run capacity constraints. I show that Ramsey prices can result in minimal deadweight loss in the short run when capacity constraints bind.

The paper proceeds as follows. Section 2 provides information on California’s fast charging market. In Section 3, I describe the data collection and panel construction process. In Section 5, I present estimates of the impact of station entry on charging quantities, price, and quality. In Section 6, I show the theoretical efficiency of markups, build and estimate a model of heterogeneous demand, and use the elasticities from the model of demand to calculate profit and social surplus maximizing prices. Finally, Section 7 concludes.

## 2 Fast Charging

Electric vehicle charging is generally classified into three levels based on speed. Level 1 and level 2 charging take 4-10 and 40-50 hours to charge a vehicle, respectively. In this paper, I focus on level 3 charging, often called DC fast charging or simply fast charging, which can refuel a car in 20 minutes.<sup>2</sup> Fast charging is generally considered the infrastructure necessary to enable longer trips and provide regular refueling to those who cannot charge at home or work.

The downside to fast charging is the cost. Chargers are expensive and often require upgrades to electricity grid infrastructure before they can be connected. Stations generally cost \$40,000 to \$400,000 per charger depending on the speed of the charger, expense of upgrading local grid infrastructure, and number of chargers at the station (Gamage, Tal and Jenn, 2023; Li, 2023; Miyatra, 2025). An additional barrier to installation is the time it takes to upgrade the grid before chargers can be connected, which can take several months (Gamage, Tal and Jenn, 2023).

### 2.1 The Fast Charging Market

The fast charging market in California is highly concentrated. Three networks, Tesla, EVgo, and Electrify America, operate 64% of stations.<sup>3</sup> Tesla, the largest of the three, builds larger stations than nearly all other networks; it operates 30% of stations and 57% of fast chargers in the state. Networks have raised consumer search costs through only listing prices on their own mobile applications. Tesla and Electrify America all share charger availability data with Google Maps, but none of the three shares prices.<sup>4</sup>

Another key way that stations have segmented the market is through differences in charging standards. DC fast chargers have been developed with three different charging standards.<sup>5</sup> All Teslas and Tesla charging stations exclusively use Tesla’s charging plug, the North American Charging System (NACS). Until 2025, no other cars came equipped for NACS plugs and instead were equipped for either CCS (also called J1772 combo) or CHAdeMO plugs. Further, other stations did not offer NACS charging until 2025, but have begun to incorporate NACS plugs in new stations and retrofit some existing stations with NACS plugs. Automakers sell adapters to allow owners to charge at stations with different

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<sup>2</sup>I provide a table with more information on the three charging levels in [Appendix Table 1](#).

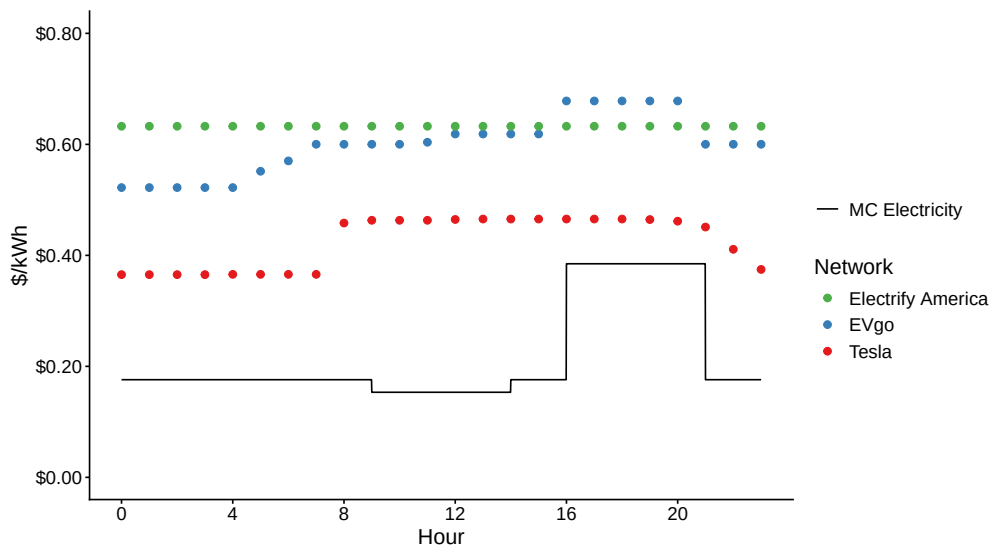
<sup>3</sup>I define stations as any set of multiple chargers operated by a network in a single location.

<sup>4</sup>Another source of charging station information is PlugShare, a web and mobile application that is the largest public aggregator of EV charger information. Neither Tesla nor Electrify America shares prices on PlugShare.

<sup>5</sup>For more on the history of charging standard differences and the consequences for EV adoption, see [Li \(2023\)](#).

charging standards, but development of NACS adapters has been slow and they did not become readily available until the second half of 2025.

Markups above marginal operating costs are high in the fast charging market. [Figure 1](#) shows networks’ average hourly prices at DCFC stations served by PG&E during October 2025. The black line on the figure represents the marginal cost of one kWh of electricity for stations on the most commonly used electricity tariffs during this period.<sup>6</sup> The quantity-weighted average markup above marginal operating cost from October 2024 to October 2025 was 109%, which is an order of magnitude greater than average markups in the retail gasoline industry.<sup>7</sup>



*Notes:* Data is for October 2025. Prices are averaged across all stations operated by each network in PG&E’s service area. “MC Electricity” is the per kWh electricity cost under the most electricity rates most frequently used by fast charging stations.

Figure 1: Average Hourly Prices and Marginal Costs at DCFC Stations in PG&E Service Area

<sup>6</sup>Smaller stations were more likely to use the lower capacity tariffs, which charged slightly higher prices per kWh that were within \$0.02/kWh of the black line at all hours.

<sup>7</sup>Providing charging does not require inputs beyond electricity, so I consider the marginal operating cost to be the cost of purchasing one kWh. There are other fixed operating and maintenance costs such as maintenance and monthly capacity fees that I defer to [Section 6](#).

## 3 Data

### 3.1 Station Characteristics

I collect a list of public fast charging stations and characteristics for those stations from the Alternative Fueling Station Location API run by the Department of Energy’s Alternative Fuels Data Center (AFDC). The data include station address, geographic location, number of fast chargers, and charging standards (NACS, CCS, ChaDeMo) available at the station. I use this station list as the foundation for building weekly panels of station prices, use, and quality.<sup>8</sup>

### 3.2 Price Data

I build a weekly panel of hourly station prices in California from October 2024 through May 2026. Each week, a team of research assistants and I collected hourly prices for all Tesla, EVgo, and Electrify America DCFC stations in California. Data comes from each company’s mobile application and PlugShare, a third party aggregator of charging station data, which posts EVgo prices. The list of stations for a week comes from querying the Alternative Fueling Station Location API for a station list each Saturday. We collected prices for Tesla, EVgo, and Electrify America stations on the station list between Saturday and the following Friday.<sup>9</sup> My recorded weekly prices at a station will not capture price changes that occurred during that week if the changes occurred after data was collected.

Networks do not necessarily offer the same price to all consumers. EVgo and Electrify America offer subscription plans that lower volumetric prices in return for a monthly fee. Tesla has only opened some of its stations to drivers of non-Teslas, but at those stations prices are 40% higher for charging non-Tesla vehicles unless the consumer subscribes to a monthly plan they offer.<sup>10</sup> I use the base cost from each network (or clarify which set of prices I am referencing). For Electrify America and EVgo, these are the prices that drivers without a paid monthly subscription pay. For Tesla, this can be viewed either as the prices for a Tesla driver or for a non-Tesla driver who purchases a monthly subscription. More than half of EVs registered in California are Teslas, so one can interpret the prices in this paper as the prices the median or modal driver in California faces. I do not include each

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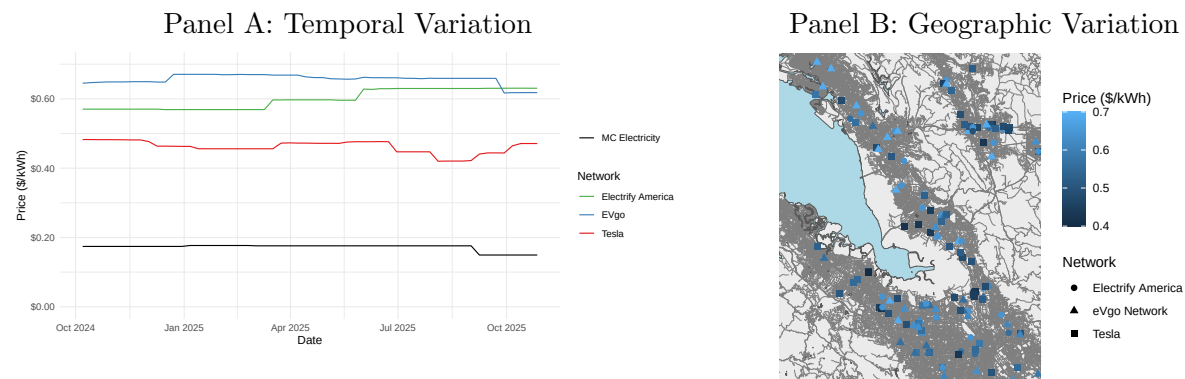
<sup>8</sup>There is no guarantee that the Alternative Fueling Station Location list contains information on every station. However, anecdotal evidence from more than a year and a half of collecting price data suggests it is highly uncommon for a station to appear on a network’s mobile application and not in the list of stations unless the station opened that week or is non-operational.

<sup>9</sup>I provide a more detailed explanation of the process in [Appendix B.1](#).

<sup>10</sup>Some Tesla drivers may also face this higher price if they purchased a used Tesla.

network’s fees for idling at the station after charging or Tesla’s congestion fee for charging above 80%.

In total, the data include 2.6 million station-week-hour prices across 1,407 charging stations. I am able to collect prices for 99.5% of Tesla, EVgo, and Electrify America stations from the AFDC’s list. [Figure 2](#) shows price variation over time and across geography. Networks change prices rarely, so most price variation comes from within-day and across-station price variation. Networks do not change prices at the same time or in the same direction, which is suggestive that pricing is not collusive. There is spatial variation in prices across a network’s stations. Tesla and EVgo predominantly used time-of-use pricing (where prices vary throughout the day) for most of the sample period while Electrify America used a single price at most stations. In November of 2025, Tesla moved to demand-based pricing (where prices vary throughout the day) at most of its stations. I cannot view hourly prices at these stations and drop all station-weeks using demand-based pricing from analysis and limit my demand estimation to the period preceeding this change.



*Notes:* Both panels show 2 PM charging prices for stations served by PG&E. Panel A presents average weekly price by network. Panel B shows average price at each of a subset of stations during October 2025.

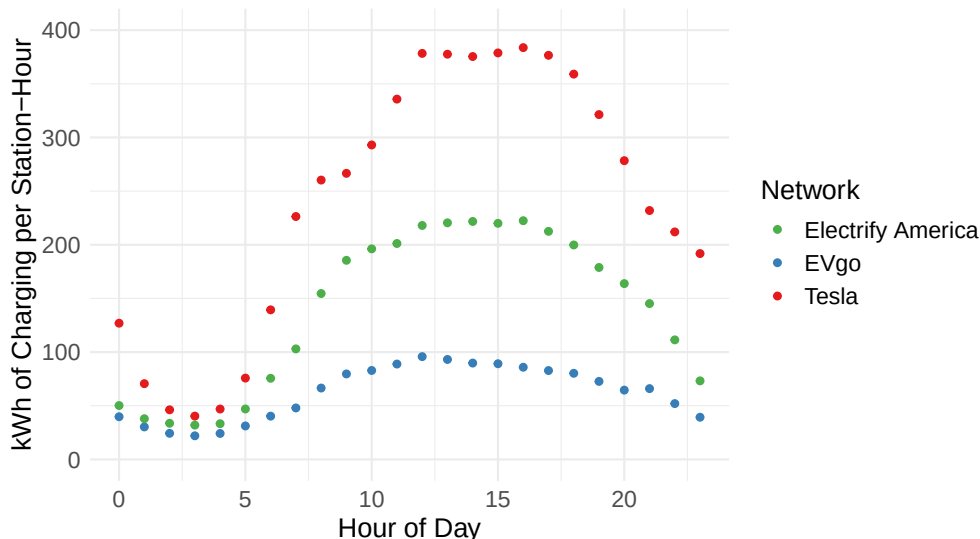
Figure 2: Variation in 2 PM Charging Prices

### 3.3 Electricity Utilization and Billing Data

I obtain station-level electricity consumption and billing data from Pacific Gas & Electric Company (PG&E), the largest electric utility in California and one of the largest in the United States. The data cover 1/1/2022-12/31/2025 for fast charging stations that PG&E identified in their data or matched to addresses from the AFDC station list. The data includes both meter-level electricity consumption measured in kWh and rate payer category.

I convert the raw electricity usage data into an approximation of the amount of electricity used for charging each hour. The raw meter data incorporates all metered electricity consumption at the location, which includes electricity for payment systems and lights. I attempt to remove non-charging electricity consumption by subtracting the lowest observed electricity use for each station. I classify the remainder as electricity used for charging.<sup>11</sup> I plot the average hourly use by network in [Figure 3](#).<sup>12</sup>

Using the rate payer category and historical rates, I can determine the marginal cost of a kWh for each station at each hour of the day. Unfortunately, I cannot view the capacity fees paid by stations each month. The PG&E data allow me to measure hourly charging quantities and marginal operating costs at 455 stations that are in the PG&E data and do not share a meter with another station.



*Notes:* Data is for October 2025. Electricity utilization is averaged across all stations operated by each network in PG&E’s service area.

Figure 3: Average Hourly Electricity Use at DCFC Stations in PG&E Service Area

### 3.4 Station Quality Data

My last data source is a measure of quality from Paren, a charging station data company. Specifically, I use Paren’s Station Health Score, a weekly index of station quality. The score

<sup>11</sup>I document this process more thoroughly in [Appendix B.2](#).

<sup>12</sup>Large differences across networks reflect differences in the number of chargers per station, speed of chargers, and customer usage. I plot use per charger (rather than per station) in [Appendix Figure 1](#), which switches the rank order of Tesla and Electrify America. This reflects that Electrify America stations are smaller than Tesla stations and Electrify America’s chargers have higher (advertised) charging speeds.

is calculated using a proprietary algorithm that incorporates a charger’s uptime and share of charging session attempts that were successful during the past week. A Station Health Score (referred to as health index for the remainder of the paper) of 100 during a week would indicate that the station was operational the whole week and all customers that attempted to initiate a charge at the station did so successfully. Paren’s data covers all EVgo and Electrify America stations throughout the sample period and the majority of Tesla stations in 2025.<sup>13</sup> As part of my data use agreement, I do not present quality results disaggregated by network.

Paren began collecting price data in 2025. I match my station-week-hour price data with their price data for September 2025 to assess the accuracy of my data. I find that more than 99% of station week-hour prices match across the two datasets, providing support for the accuracy of my price data.

Table 1: Summary Statistics

|                             | Tesla  | EVgo   | Electrify<br>America | Other  |
|-----------------------------|--------|--------|----------------------|--------|
| <b>All California</b>       |        |        |                      |        |
| Stations                    | 663    | 454    | 296                  | 1014   |
| Plugs per Station           | 16.4   | 4.9    | 4.9                  | 5.2    |
| Price                       | \$0.43 | \$0.59 | \$0.61               |        |
| Health Score                | 96.0   | 80.0   | 92.8                 | 74.7   |
| Entered 2025 or later       | 21%    | 14%    | 6%                   | 30%    |
| Stations within 5 min.      | 2.4    | 2.7    | 2.1                  | 2.6    |
| <b>Electricity Use Data</b> |        |        |                      |        |
| Stations                    | 178    | 125    | 90                   | 62     |
| Plugs per Station           | 15.1   | 5.5    | 4.5                  | 9.0    |
| Price                       | \$0.44 | \$0.58 | \$0.61               |        |
| Health Score                | 97.4   | 81.6   | 91.5                 | 82.2   |
| Daily kWh                   | 5761.0 | 1276.5 | 2478.7               | 1060.2 |
| Stations within 5 min.      | 2.1    | 3.0    | 1.9                  | 2.9    |

## 4 Markups

### 4.1 Product Differentiation

In Bertrand competition, product differentiation allows firms to have price setting power. The more differentiated products are, the less each firm must consider the actions of the others. In the extreme case, consumers do not substitute between goods and firms act as monopolists. At the other extreme, when products are perfect substitutes, firms are

<sup>13</sup>Paren did not initially have good coverage of Tesla stations, but expanded their Tesla station data in 2025.

perfectly competitive and cannot charge prices above marginal cost. Fast charging stations choose prices rather than quantities, so the Bertrand model fits my setting well. [Perloff and Salop \(1985\)](#) provide a helpful model for understanding the components that may explain the magnitude of markups in the fast charging market.

Consider a consumer that must choose where to charge among  $n$  stations indexed by  $i$ . The consumer's preference for charging at each station is given by the vector  $\boldsymbol{\theta} = (\theta_1, \dots, \theta_n)$  where  $\theta_i$  includes preferences over nearby amenities (such as grocery stores, restaurants, etc.), charging network, charging standard, and travel costs. The vector of prices at each station is  $\mathbf{p} = (p_1, \dots, p_n)$ . The consumer chooses to charge at the station that maximizes net surplus  $s_i = \theta_i - p_i$  for  $i \in 1, \dots, n$ .<sup>14</sup>

For simplicity, let all consumers' preferences for stations be independent and identically distributed. Each density function is given by  $g(\theta) = g(\theta_i)$  with cumulative distribution function  $G(\theta)$ . The probability that  $s_i \geq s_j$  is  $\Pr(s_i \geq s_j) = \Pr(\theta_j \leq \theta_i + p_j - p_i) = G(\theta_i + p_j - p_i)$ . Aggregating over  $L$  consumers who each charge at the station that gives them the greatest surplus gives that station  $i$  supplies quantity:

$$Q_i(\mathbf{p}) = \Pr(s_i \geq \max_{j \neq i} s_j)L = L \int \Pi_{j \neq i} [G(\theta_i + p_j - p_i)] g(\theta_i) d\theta_i. \quad (1)$$

Let each station face constant marginal cost  $c$ . Then its profit function is:

$$\pi_i(\mathbf{p}) = (p_i - c)Q_i(\mathbf{p}) - K \quad (2)$$

where  $K$  is the station's fixed cost of entering. Holding competitors' prices constant, the profit maximizing price for each station is:

$$p_i = c - \frac{Q_i(\mathbf{p})}{\partial Q_i / \partial p_i}. \quad (3)$$

Assuming symmetry, [Perloff and Salop \(1985\)](#) find a Nash-Bertrand equilibrium where all stations have the same quantity  $Q = \frac{L}{n}$  and charge the same price

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<sup>14</sup>For simplicity, I have omitted the outside option of not charging at any station. Adding in the outside option sets an additional restriction that  $s_i \geq \bar{v}$  for the consumer to choose station  $i$ .

$$p(n) = c - \frac{Q_i(\mathbf{p})}{\partial Q / \partial p} = c + \frac{1}{M(n)} \quad (4)$$

where  $M(n) = n(n-1) \int [G(\theta)]^{n-2} [g(\theta)]^2 d\theta$ .  $M(n)$  is the slope of the demand curve multiplied by the station's market share,  $\frac{n}{L}$ . That is, the markup above  $c$  is proportional to the number of stations in the market.

The model has four key implications for the fast charging market. First, product differentiation sustains an equilibrium with prices greater than marginal cost. Second, preference strength impacts equilibrium markups. To see this, let  $\theta = \beta\theta_0$  where  $\beta$  is a measure of preference strength and  $\theta$  is the draw from a mean zero distribution. Then equilibrium  $p(n) = c + \frac{\beta}{M(n)}$  and the markup is proportional to preference strength. As preferences for charging networks or station amenities increase, equilibrium markups also increase.

The third implication is that station entry increases competition and reduces prices and markups. As  $n$  increases,  $M(n)$  increases, the markup decreases, and  $p$  decreases. The fourth is that fixed costs affect the entry decision, but not a station's pricing choice once it has entered. If a network is considering building a station, it needs to expect to earn  $p = (p(n) - c)Q - K = (p(n) - c)\frac{L}{n} - K \geq 0$ . Network would choose to continue to build new stations until  $p(n) = \frac{nK}{L} + c$ . So even though fixed costs do not enter the optimal price, they do affect how many stations will be built and equilibrium markups and prices.

## 5 Station Entry

The model in [Section 4.1](#) shows that new station entry should incumbent stations to lower prices. Moreover, station entry provides a natural experiment to test for price competition. If stations respond to competitor entry by lowering prices, this aligns with the model of Bertrand competition amongst partially differentiated stations. If stations do not respond by lowering prices, this would suggest that networks are not profit maximizing, that they compete on a measure other than price, or that stations are highly differentiated. Such a result may be due to market frictions – [Arcidiacono et al. \(2020\)](#) find supply-side frictions that lead grocery stores to not update prices. On the demand side, high search costs could deter competition ([Satterthwaite, 1979](#)).

## 5.1 Empirical Design

I consider the effect of station entry on incumbent charging quantities, prices, and quality. I limit my sample to incumbent stations within a five-minute drive of the entrant to focus on dynamics between stations that one would expect to be substitutes.<sup>15</sup> I measure responses to market entry by exploiting the exact timing of entry. Networks choose new locations based on market conditions, but the licensing and infrastructure upgrade process takes several months and is subject to a large amount of variation in how long it takes. Once operational, stations have no reason to delay opening as operational costs are below their expected revenues. This means that the exact timing of entry is determined by administrative and technical factors rather than market conditions. My identifying assumption, that stations do not control the exact timing of entry, is the same that [Arcidiacono et al. \(2020\)](#), [Davis, McRae and Seira \(2025\)](#), [Taylor and Muehlegger \(2025\)](#), and others use.

My primary empirical approach is a difference-in-differences event study model. Station construction is public, so a key threat to causal identification is anticipation. If stations lower prices in anticipation of station entry, then a standard difference-in-differences approach that averages pre-period prices would underestimate the counterfactual price in the absence of entry and bias the difference-in-differences estimate. I use an event study specification to show trends during the twelve weeks prior to station opening and disaggregate results by the number of weeks from treatment. This approach also has the benefit of showing how long it takes for quantities and prices to adjust.

I match treated stations to untreated stations to create a counterfactual for outcomes in the absence of openings. I designate untreated stations as those that did not have an opening within a 10-minute drive within 24 weeks of the treatment opening. I limit matching to untreated stations operated by the same charging network, in the same core-based statistical area (CBSA), and with a similar number of nearby stations.<sup>16</sup> The goal of this approach is to compare treated stations to stations that were affected by the same market trends and faced a similar amount of competition prior to the entry event. The  $k$  counterfactual stations for each treated station are given a weight of  $\frac{1}{k}$  to ensure each treatment event is weighted equally.

I provide a visual representation of this process in [Figure 4](#). In this example, a new Tesla station (red circle) opens within a five-minute drive (depicted by the red line) of a Tesla (green circle), Electrify America (blue square), and EVgo (orange triangle) station. Each

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<sup>15</sup>I explain the role of distance in [Section 5.2.6](#) and provide results using a 10-minute threshold in the appendix.

<sup>16</sup>I require the number of nearby stations at matched stations to be within one when the treated station has four or fewer nearby stations, within two when the treated station has five to nine nearby stations, or within three when the treated station has 10 or more nearby stations.

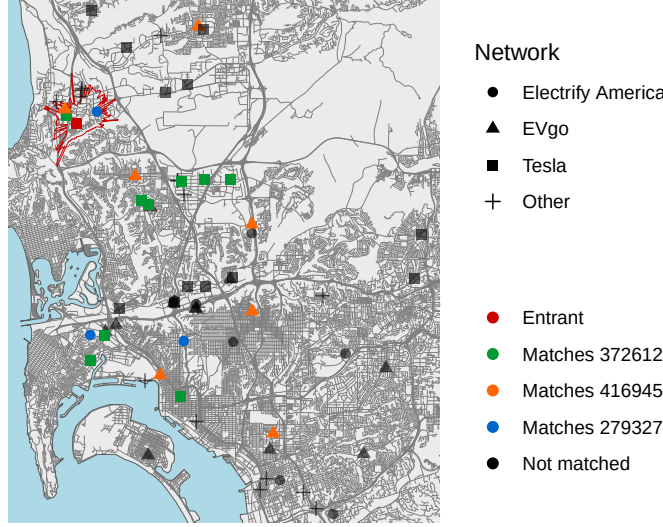


Figure 4: Visual Example of Matching Process

of those stations is matched to all the corresponding stations that are the same color and shape as the treated station. Those stations are all part of the same charging network, in the San Diego-Carlsbad CBSA, and near a similar (within one) number of stations. The map illustrates why I require nearly the same number of nearby stations for control stations. When electricity rates fall and marginal costs decrease, EVgo might lower prices at the orange triangle stations but leave prices unchanged where it has little competition, such as its station on Coronado Island, a black triangle towards the bottom left of the figure.

The event study runs from twelve weeks before entry to twelve weeks after entry. I avoid overlapping treatment by restricting to only entrances near an incumbent without another nearby entry event within 24 weeks. I consider incumbent electricity use, price, and health index as outcomes and run regressions separately for each network for quantity and price in case driver behavior and incumbent strategies vary by network.<sup>17</sup> I estimate the equation:

$$y_{ihw} = \sum_{\tau=-12}^{\tau=12} \beta_{\tau d} \mathbf{1}[\text{EventWeek}_{iw} = \tau] \times \mathbf{1}[\text{Treated}_i] + \mu_{ih} + \gamma_{n(i)w} + \lambda_{u(i)m(w)} + \varepsilon_{ihw} \quad (5)$$

where  $\tau$  represents event week and  $\mathbf{1}[\text{Treated}_i]$  is a dummy variable equal to one for incumbents exposed to entrants and zero for the counterfactual group. I estimate station price and usage at the station-week-hour level to account for different prices and charging over the course of the day. I include station-hour fixed effects,  $\mu_{ih}$ , to control for time-invariant sta-

<sup>17</sup>For regressions with health index as the outcome variable, I present results aggregated across all networks due to my data sharing agreement.

tion characteristics, network-week fixed effects,  $\gamma_{n(i)w}$ , to control for network pricing behavior over time, and utility-month of sample fixed effects,  $\lambda_{u(i)m(w)}$ , to capture changes in marginal costs over time. I estimate the impacts of opening on health index at the station-week level and therefore include station fixed effects instead of station-hour fixed effects.

I apply two-way clustered standard errors at the CBSA and year-month level, because the probability of treatment varies across CBSAs and time. I do not interact CBSA with year-month because the error for one CBSA in a given month is likely correlated with the error the previous month (Colin Cameron and Miller, 2015). The total number of clusters ranges from 51 to 59 across analyses.

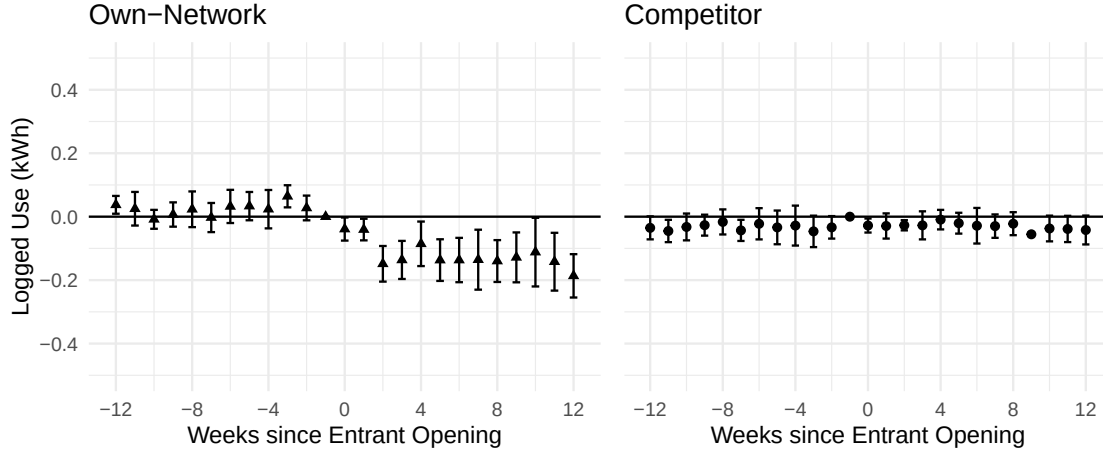
I elect to include already treated stations in my analysis. This is because most stations that have a nearby competitor have already been treated (even if treatment was before the study period) and I want to understand the effect in both areas where there are already multiple stations (in addition to those with few stations). I set a 24 week threshold between entry events to ensure that no station-week in my analysis is treated by multiple entries. In these cases, I drop all entry events to ensure the pre- and post-periods are not contaminated. Appendix Figure 3 shows that limiting to just the first entry would cut my sample size by 25% and that station exit is incredibly rare during this time.

In addition to the event study models, I also estimate a standard difference-in-differences model. This model uses the same data but groups the twelve pre-entry weeks into a single pre-period group and the week of entry and following twelve weeks into a single post-period group. The difference-in-differences estimator recovers the change in the difference between the treatment and control group from before to after entry.

## 5.2 Results

### 5.2.1 Charging Quantities

Market entry has asymmetric effects on own- and competing-network stations. Figure 5 shows that charging at existing stations decreased following own-network entry but remains unchanged following competing network entry. The decrease from own-network entry occurs shortly after station entry and is sustained for the event study’s twelve week window at around a 14% decrease from just before entry. Appendix Figure 4 shows the own-network effect is driven by Tesla stations and Appendix Figure 5 shows the effect of competitor entry on charging is null across all networks. I provide results from the difference-in-differences model for these and the following event study analyses in Table 2.

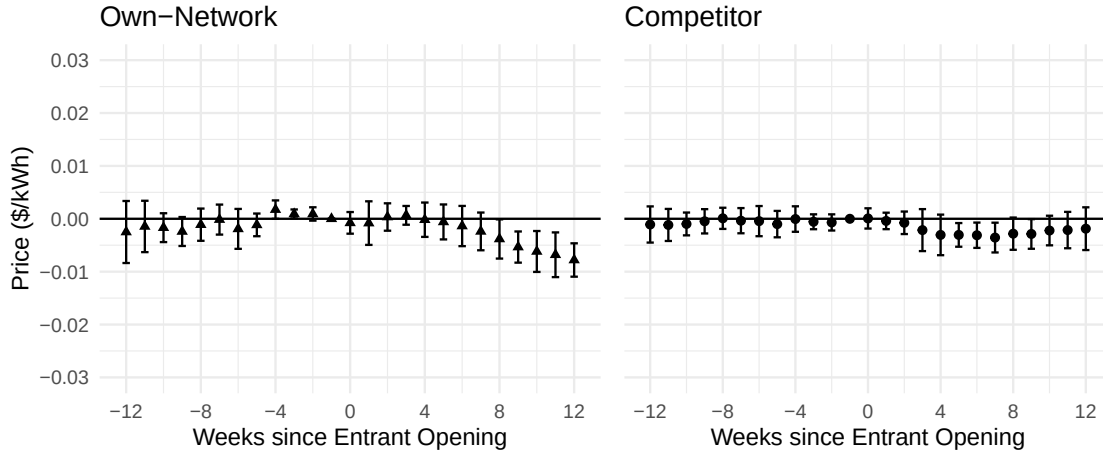


*Notes:* Data is for total electricity consumption at each station and spans January 2022 - December 2025.

Figure 5: Event Study Estimates of Own-Network Station Entry within Five Minutes on Incumbent Hourly Electricity Consumption

### 5.2.2 Prices

Price responses generally align with changes in charging volumes. [Figure 6](#) shows no anticipatory price changes in the twelve weeks before a station's opening. After station entry, prices fall slightly at own-network stations but not competing stations. Again, the own-network effect comes from Tesla stations and the non-response to competitor entry is consistent across networks (see [Appendix Figure 6](#) and [Appendix Figure 7](#)).



*Notes:* Data is for station-week-hour prices and spans October 2024 - May 2026.

Figure 6: Event Study Estimates of Own-Network Station Entry within Five Minutes on Incumbent Price

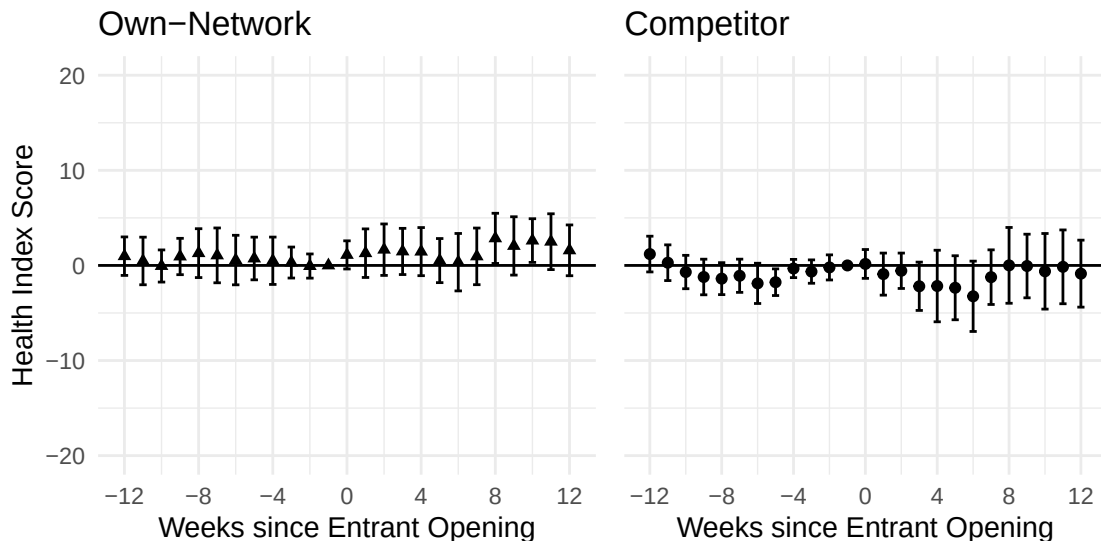
The use results showed that Tesla’s new stations cannibalize some demand from any nearby incumbents, shifting demand down and leading to a lower optimal price. The gradual decrease does not reflect Tesla changing prices repeatedly, but rather that Tesla rarely changes prices, so the price change may lag the demand change by a couple weeks or months depending on when the network next updates prices. Similarly, stations experience no decrease in demand following competitor entry and therefore have no incentive to lower prices. The difference-in-differences results in [Table 2](#) suggest I can rule out average competitor price responses greater than 0.4 cents and network-specific prices responses greater than 0.7 cents. Using the average price and marginal electricity cost for the stations in the PG&E data, the confidence interval bound of 0.4 cents maps to 1.5% of price minus marginal operating cost margins. Therefore I can rule out price decreases in the realm of the 6-7% estimated for the retail gasoline industry ([Taylor and Muehlegger, 2025](#); [Davis, McRae and Seira, 2025](#)).

### 5.2.3 Quality

I also find no evidence that incumbent stations improve quality in response to greater competition. [Figure 7](#) shows the effect of station entry on own-network and competing station health index score. Taken together, the price and quality results show stations do not appear to respond to competitor entry. This is a rational decision – I detect no change in demand for stations following competitor entry, so the optimal prices and quality after entry should be the same as prior to entry. It is still possible that stations compete on other measures, or that competition is at the network level – networks make decisions on prices and quality that affect all stations in an area at once. I explore network-level pricing in [Section 6](#).

### 5.2.4 Charging Standards

One possible explanation for why I detect no cross-network substitution is that different networks use different charging standards. Almost all cars are only equipped to charge with one charging standard, so drivers must either go to a station with that standard or own the requisite adapter. Prior to 2025, all non-Tesla stations used CCS, CHAdeMO, or both, while Tesla exclusively used NACS. Since 2025, non-Tesla networks have begun to build new stations with NACS plugs and retrofit old stations to include NACS plugs (though the latter is not common). Non-Tesla manufacturers also began to make NACS adapters available to drivers in 2025, though they were in short supply at the start of the year. My use data cover 2022 through 2025, so a preference for stations with NACS chargers could appear as a network preference.

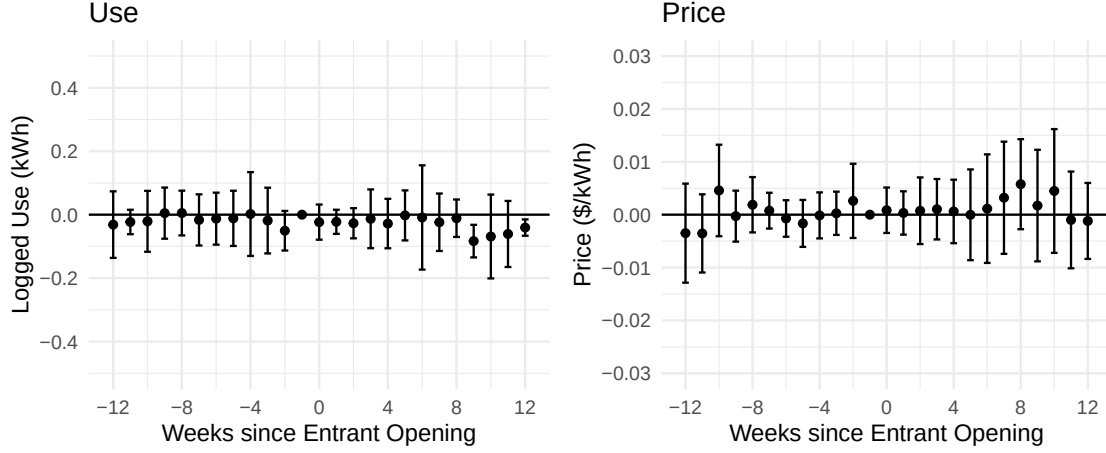


*Notes:* Data is for station-week health index score and spans October 2024 - May 2026.

Figure 7: Event Study Estimates of Station Entry within Five Minutes on Station Quality

I use three approaches to evaluate whether the lack of substitution across networks reflects preferences for charging standards rather than networks. First, I run the same usage event study for competitor entry as in [Section 5.2.1](#) but limit to just entrants using the same charging standard as the incumbent. The changes in charging standards and converter accessibility in 2025 motivate two additional experiments. The first is to limit the window of analysis to 2025 onwards. If non-Tesla networks begin competing with Tesla for customers in 2025 now that more non-Tesla stations had NACS plugs and more non-Tesla drivers had NACS adapters, focusing on just the period with competition should reveal an impact of competitor station openings. Second, my weekly panel of station information allows me to determine when networks retrofitted existing stations to add NACS chargers. I use the same event study approach to determine whether charging volumes or prices at nearby existing Tesla stations change following addition of NACS chargers.

[Figure 8](#) shows that there is no substitution across stations from different networks that have the same charging standard and stations still do not lower prices. I provided results by network for use and price in [Appendix Figure 8](#) and [Appendix Figure 9](#), respectively. Additionally, there is no substitution from Tesla stations to non-Tesla stations when the non-Tesla stations add NACS chargers (see [Appendix Figure 10](#)) or to new non-Tesla stations in 2025 when these stations are potentially better substitutes (see [Appendix Figure 11](#)). These results suggest it is network rather than charging standard differences that drive the lack of substitution across networks. One potential explanation is that drivers do not consider all possible stations with the charging plug type they need, but instead open one network's



*Notes:* Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Figure 8: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Five Minutes

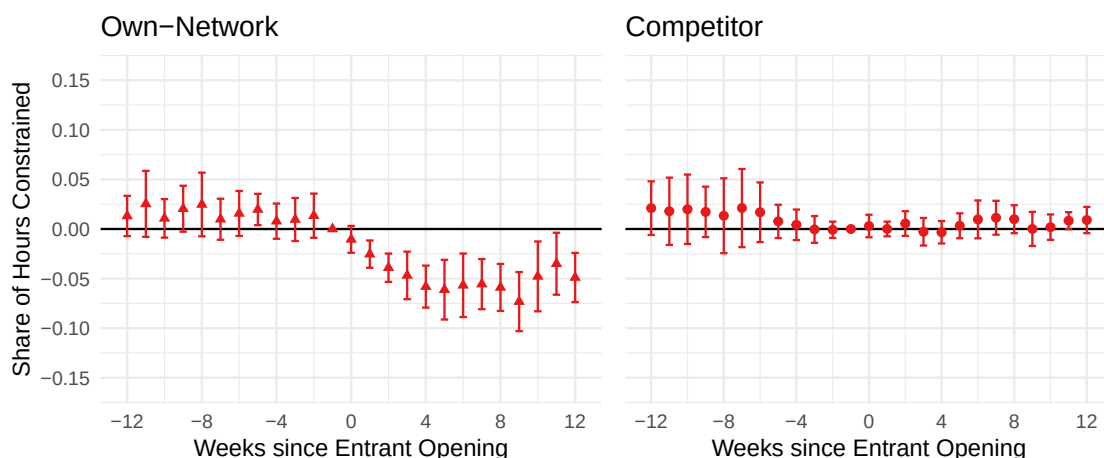
application to look for nearby stations.

### 5.2.5 Capacity

A notable difference between the retail gasoline literature and my results is that Tesla lowers their prices after they open a station near one of their own stations. One potential explanation is that the new station cannibalizes demand, leading to a downward shift in demand at the incumbent stations. This shift leads to a decrease in optimal price. It is also possible that capacity constraints explain part of the price effect.

Stations are limited in how many cars they can charge and how much electricity they can discharge at once. When all chargers at a station are full, the marginal cost curve becomes vertical as the station would need to add another charger to increase quantity, but cannot do so in the short run. Even when the station is not completely full, an additional vehicle may slow charging speeds (because technical constraints restrict the maximum electricity a station can discharge) or deter potential customers from driving to the station if they use a mobile application to check how full the station is before driving there. This could be modeled as an increase in the marginal cost of providing one kWh above the marginal operating cost. If Tesla entrants relieve capacity constraints, this would reduce marginal costs at the incumbent station by eliminating periods where capacity constraints increase marginal costs. This may explain why prices fall at Tesla stations following nearby Tesla entry and why the network chooses to build stations that will cannibalize charging from existing stations.

I use the same event study approach from the prior entry analyses to test whether entry relieves capacity constraints. Ideally I would define a station as capacity constrained during any hour where the station must reduce the amount of electricity drawn by each charger or there are ever no chargers are available. I cannot observe either in my data, so I use a rough measure of how close electricity discharge is during an hour to the maximum amount of electricity the station can discharge. I calculate the observed electricity usage during any given hour during the station's history and designate an hour as potentially capacity constrained (at or near capacity) if electricity consumption during that hour is at least 80% of a benchmark for the maximum feasible amount of electricity consumption. I set that benchmark as each station's 98<sup>th</sup> percentile of electricity consumption during daytime charging hours.<sup>18</sup> Using this approach, I classify 5.4% of station hours as capacity constrained.



*Notes:* Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Figure 9: Event Study Estimates of Station Entry within Five Minutes of Tesla Station on Share of Hours near Charging Capacity

The events study results in [Figure 9](#) show that Tesla stations exposed to new own-network stations significantly reduce the likelihood of capacity constraints. The probability of a Tesla station being potentially capacity constrained decreases by 6.6 percentage points. This estimate is very large in magnitude – a 47% decrease from the 14 percent of pre-period hours classified as potentially capacity constrained. This suggests stations may lower prices following nearby own-network entry because they are less likely to be capacity constrained.

<sup>18</sup>Charging speeds vary by car model and the car battery's charge level (charging gets notably slower once the battery is more than 80% full – Electrify America doesn't allow charging past 85% at some stations and Tesla charges extra for cars charging above 80%) so there is no set amount of electricity a fully occupied station would discharge. I therefore use the 98<sup>th</sup> percentile of charging from 8 AM to 8 PM to rule out hours where cars were exceptionally fast in terms of charging speeds. Changing this threshold up or down makes

Table 2: Difference-in-Differences Results

|                                              | All        |           | Tesla      |          | EVgo       |         | Electrify America |
|----------------------------------------------|------------|-----------|------------|----------|------------|---------|-------------------|
|                                              | Competitor | Own       | Competitor | Own      | Competitor | Own     | Competitor        |
| <b>Log Use (kWh/hour)</b>                    |            |           |            |          |            |         |                   |
| Estimate                                     | 0.000      | -0.143**  | 0.001      | -0.179** | 0.014      | -0.073* | -0.014            |
|                                              | (0.014)    | (0.019)   | (0.017)    | (0.028)  | (0.025)    | (0.013) | (0.036)           |
| Baseline mean                                | 4.557      | 4.652     | 5.371      | 5.457    | 3.624      | 3.274   | 4.294             |
| Observations                                 | 1,309,200  | 335,400   | 478,200    | 147,000  | 420,600    | 117,000 | 367,200           |
| <b>Price (\$/kWh)</b>                        |            |           |            |          |            |         |                   |
| Estimate                                     | -0.002     | -0.002    | -0.002     | -0.005*  | -0.001     | 0.006** | -0.001            |
|                                              | (0.001)    | (0.002)   | (0.004)    | (0.002)  | (0.001)    | (0.001) | (0.003)           |
| Baseline mean                                | 0.554      | 0.468     | 0.415      | 0.424    | 0.604      | 0.590   | 0.612             |
| Observations                                 | 5,108,616  | 2,007,072 | 1,512,216  | 745,056  | 2,371,512  | 854,928 | 1,224,888         |
| <b>Health Index</b>                          |            |           |            |          |            |         |                   |
| Estimate                                     | -0.463     | 0.948     |            |          |            |         |                   |
|                                              | (1.553)    | (0.959)   |            |          |            |         |                   |
| Baseline mean                                | 88.6       | 93.1      |            |          |            |         |                   |
| Observations                                 | 2,998,560  | 848,208   |            |          |            |         |                   |
| <b>Share Hours with Constrained Capacity</b> |            |           |            |          |            |         |                   |
| Estimate                                     | -0.004     | -0.046**  | -0.007     | -0.061** | -0.002     | -0.013  | -0.001            |
|                                              | (0.007)    | (0.008)   | (0.008)    | (0.005)  | (0.003)    | (0.004) | (0.008)           |
| Baseline mean                                | 0.079      | 0.109     | 0.096      | 0.126    | 0.052      | 0.073   | 0.080             |
| Observations                                 | 1,309,200  | 335,400   | 478,200    | 147,000  | 420,600    | 117,000 | 367,200           |

Notes: \*/ \*\* Denote significance at  $p < 0.05/0.01$  level.

## 5.2.6 Distance

In the previous models, I limit treatment to the effect of openings within five minutes and compared to similar stations without any openings within ten minutes. In this section, I test this assumption on the distance over which consumers are willing to substitute between stations. I compare stations treated by openings that are 0-5 minutes, 5-10 minutes, and 10-15 minutes away to control stations without any openings within a 15-minute drive. I focus specifically on the effect of Tesla openings on Tesla incumbents as this combination of existing and new stations was the one with the largest magnitude estimates.

I increase sample size for the analysis of the effect on charging quantities by shrinking the analysis window to eight weeks before and after opening. The results in [Figure 5](#) show that the effect of entry on charging quantities is stable after two weeks, so changing the window should not be a concern for the parallel trends requirement. The effect on prices in [Figure 6](#) doesn't materialize until six to eight weeks after opening and continues to grow through the study period, so I do not change the analysis window for the price analysis.

I present results for the effects of own-network openings on incumbent Tesla stations in [Appendix Figure 12](#). The effect of new Tesla stations on nearby existing Tesla station

a slight difference in the share of hours classified as at capacity, but no difference in the relative effect size.

charging quantities is largest for the 0-5 minute treatment group, but also negative and significant for the 5-10 minute treatment group. Even though there is no effect on charging for stations 10-15 minutes away, all three groups decrease prices following entry. This could reflect that rather than lowering prices in response to decreased demand, Tesla is decreasing prices in anticipation of a demand response, but it doesn't materialize for the stations 10-15 minutes away. Because my results suggest that drivers are willing to switch to new stations 5-10 minutes away from an existing station, I conduct all event studies in this section using a 10-minute treatment window and provide those results in the appendix (see [Appendix Figure 13-Appendix Figure 17](#)).

## 6 Pricing Strategy and Efficiency

### 6.1 Pricing Efficiency

Descriptive evidence in [Figure 1](#) shows large markups above marginal operating costs, and the empirical results in [Section 5.2](#) suggest increasing the number of stations in the market does not decrease markups. However, markups are not always inefficient in the short run. When supply is constrained, raising the price to the market clearing price ensures efficient allocation. High fixed costs deter entry and preserve markup power, which allows firms to recover fixed costs. Without markups, firms would choose not to enter (assuming non-increasing average variable costs). This section will explore the effect of various pricing strategies on social surplus in the fast charging market and estimate the relative efficiency of observed prices.

#### 6.1.1 Capacity Constraints

If all chargers at a station are full, customers are unable to charge. This is a serious concern for networks. Even though none of the three charging networks shares its prices on Google Maps, both Tesla and Electrify America share the number of available chargers, and all three show this information in their own mobile applications. Further, Tesla switched to demand-based pricing at most of its California stations in November of 2025. Under the new pricing system, prices fluctuate based on the amount of available chargers at a station rather than time of day.<sup>19</sup>

A challenge facing charging stations is that they simultaneously have low utilization and, as shown in [Section 5.2.5](#), face capacity constraints. In [Figure 3](#), I show that demand fluc-

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<sup>19</sup>Under the new pricing system (called “live pricing”), the amount of available chargers determines which of three predetermined prices is the price for a consumer who begins charging.

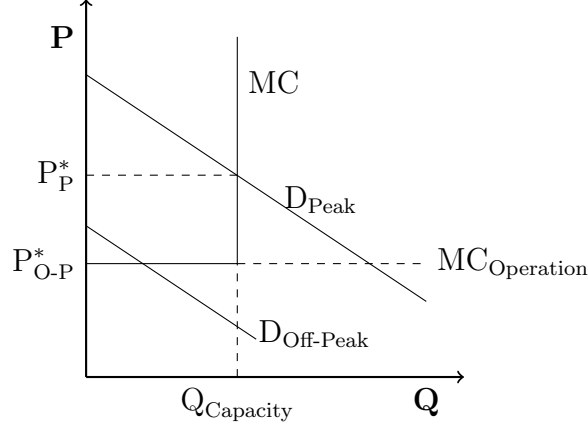


Figure 10: Short Run Efficient Pricing with Capacity Constraints

tuates periodically. It is much higher in the afternoon than during the rest of the day. Increasing peak-period capacity through building more or larger stations is costly. Chargers are expensive and marginal chargers will likely sit idle most of the day as there is already excess capacity during off-peak times, resulting in decreasing marginal revenue from additional chargers. In the long run, this could lead to an equilibrium with shortages during peak times. In this paper, I focus on the short run rather than long run and consider how networks set prices when they cannot increase charging capacity.

In the short run networks cannot increase their supply; building new stations or expanding existing stations takes months. At a station's maximum capacity, the marginal cost becomes undefined and can be drawn as a vertical line. In Figure 10, I illustrate the case where a station faces a peak-demand that is greater than off-peak demand. Short run efficient prices are those at the intersection of the demand and marginal cost curves.<sup>20</sup>

Off-peak demand intersects the marginal cost curve where  $MC(Q) = MC_{\text{Operation}}$ , and the efficient price is  $P_{\text{O-P}}^* = MC_{\text{Operation}}$ . Peak demand intersects the marginal cost curve where it is vertical. Charging a price of  $MC_{\text{Operation}}$  would lead to a excess demand, and there is no guarantee that charging will be allocated to the consumers who receive the greatest benefit. Setting the price at  $P_P^*$  equates quantity demanded and quantity supplied, ensuring

<sup>20</sup>Readers may have noticed I have dropped external costs and moved from social marginal cost to private marginal cost. Borenstein and Bushnell (2022) focus on the importance of considering external costs in determining efficient electricity rates. They show that electricity rates in California are well above social marginal cost, which means the private cost that charging stations face is greater than the social marginal cost of producing that electricity. Unless there are notable external costs from the activity of charging a car, the social marginal operating cost of charging will be below the private marginal operating cost that stations face. Therefore,  $P_{\text{O-P}}^*$  does not maximize social surplus, but rather maximizes surplus contingent on the restriction that the charging network not take losses on each unit of charging. Truly maximizing social surplus with this condition would require either PG&E to lower the per-unit cost of electricity or for the state to pay per kWh subsidies to charging stations.

that charging is allocated to consumers with the highest willingness to pay so there is no deadweight loss from allocative inefficiency. Comparing consumption and marginal operating costs over the course of the day in [Figure 3](#) and [Figure 1](#) shows that markups over marginal operating costs are highest during the early afternoon when demand peaks.

### 6.1.2 Fixed-Cost Recovery

Peak ( $P_P^*$ ) and off-peak prices ( $P_{O-P}^*$ ) in [Figure 10](#) are efficient in the short run, but do not ensure that charging networks recover fixed costs. Prices may need to be higher to incentivize the network to build a station. Raising prices above marginal costs can create deadweight loss by reducing transactions below the social surplus maximizing quantity. A common approach for regulated utilities to recover fixed costs is to charge the average cost (plus a small markup) rather than marginal cost. [Boiteux \(1956\)](#) builds on [Ramsey \(1927\)](#) to show that it is more efficient to account for heterogeneous demand elasticities. Charging larger markups on consumers who are less willing to reduce consumption when prices rise can reduce deadweight loss relative to charging the same markup to all consumers. The set of linear prices that minimize deadweight loss while recovering fixed costs, often called “Ramsey prices,” sets markups proportional to elasticities of demand across consumers or products ([Baumol and Bradford, 1970](#)).

The standard approach to calculating Ramsey prices assumes quantity can be increased at a finite marginal cost. To account for capacity constraints, I modify the Ramsey problem formulation from [Crew, Fernando and Kleindorfer \(1995\)](#) to include a condition that quantity demanded must be at or below the station’s capacity,  $\bar{X}$ . I simplify this problem by writing it for a single charging station, but the network’s problem is the same just with more products.

During a day, a charging station supplies charging at a set of times (i.e. morning, afternoon, evening)  $t \in 1, 2, \dots, T$  with constant marginal cost for each given by  $c_t$ . Heterogeneous consumers each have demand for charging at different time periods,  $\mathbf{x} = (x_1, \dots, x_T)$ , based on their individual valuation of consumption bundles,  $V(\mathbf{x}, i)$ , where  $i$  indexes consumers.<sup>21</sup> The total demand for charging at time period  $t$  is the sum of individual demand,  $x_t(\mathbf{P}, i)$ , which is a function of all time period prices and individual preferences:

$$X_t(\mathbf{P}) = \int_I x_t(\mathbf{P}, i) di, \quad t \in 1, \dots, T. \quad (6)$$

The constrained Ramsey problem is to choose the set of prices,  $\mathbf{P}$ , that maximize social surplus,  $W(\mathbf{P})$ , subject to capacity constraints and recovering fixed costs. I write this as:

---

<sup>21</sup>[Crew, Fernando and Kleindorfer \(1995\)](#) start from the Hicksian demand function to develop  $V(\mathbf{x}, i)$ .

$$\begin{aligned}
\max_{\mathbf{P}} W(\mathbf{P}) &= \int_I \left[ V(\mathbf{x}, i) - \sum_t P_t x_t(\mathbf{P}, i) \right] di + \sum_t (P_t - c_t) X_t(\mathbf{P}) \\
\text{s.t. } \Pi(\mathbf{P}) &= \sum_t (P_t - c_t) X_t(\mathbf{P}) \geq \Pi_0 \\
0 &\leq X_t(\mathbf{P}) \leq \bar{X}, \quad \forall t \in 1, \dots, T
\end{aligned} \tag{7}$$

where the two added terms on the first line represent consumer surplus and station surplus, respectively. The two constraint conditions require that the station earn surplus to cover fixed costs and that the amount of charging cannot exceed capacity at any hour. Converting this maximization problem from an individual station to the network's price setting decision requires optimizing across all stations and times of day ( $t$  indexes station-times rather than just times at a single station) but otherwise does not change the problem.

The constrained Ramsey problem is very similar to both a social surplus maximization problem and the standard Ramsey problem (which I will refer to as the “unconstrained Ramsey” problem). Removing the profit restriction makes the problem analogous to the two-period problem depicted in [Figure 10](#), where  $P_P^*$  and  $P_{O-P}^*$  are efficient. Alternatively, in the absence of the capacity constraint, the problem reduces to the unconstrained Ramsey problem. This means that if the quantities demanded at the unconstrained Ramsey prices are below the capacity constraint, the unconstrained Ramsey prices are the social surplus maximizing linear prices. However, in the case where the quantity demanded at period  $t$  at the unconstrained Ramsey prices,  $\mathbf{P}^R$ , exceeds capacity, the station needs to raise the price  $P_t$  such that  $X_t(\mathbf{P}) = \bar{X}$ . If the station does not, then it will induce deadweight loss by misallocating charging to consumers without the highest willingness to pay. Additionally, raising  $P_t$  so that  $X_t = \bar{X}$  generates revenue (without increasing deadweight loss), and lowers the optimal markup over marginal operating cost on other time periods relative to the Ramsey price.

There are three possible forms for the solution to [Equation \(8\)](#). First, there is the case where the capacity constraint binds but the profit constraint does not. In all periods where demand intersects marginal cost on the vertical portion ( $X_t(\mathbf{P}^R) > \bar{X}$ ), the station generates surplus of  $(P_t - c_t)\bar{X}$ , which sum to a value greater than  $\Pi_0$ . By raising prices to eliminate deadweight loss from misallocation, the network generates sufficient surplus to recover fixed costs and does not need to raise prices above the efficient prices. Second, if the efficient prices do not generate enough surplus and the capacity constraints do not bind on the unconstrained Ramsey prices,  $\mathbf{P}^R$ , then  $\mathbf{P}^* = \mathbf{P}^R$ . Third, if both constraints bind, then the solution is to use the elasticities of demand as in unconstrained Ramsey pricing to raise prices beyond

those that solved the second case and generate surplus equal to  $\Pi_0 - \sum_{X_t=\bar{X}} (P_t - c_t) \bar{X}$ , which is the amount of surplus still required to meet the profit target. In this case, setting prices to allocate charging efficiently recovers some, but not all of, the fixed costs. The remaining fixed costs are recovered in the way that generates the least deadweight loss following the Ramsey principle of larger markups on more inelastic demand.

A limitation of my data is that there is no obvious way to define  $\bar{X}$ . In [Section 5.2.5](#), I defined a station’s maximum feasible draw as the 98<sup>th</sup> percentile of hourly daytime electricity consumption. Setting  $\bar{X}$  at the capacity multiplied by the number of hours (i.e. four afternoon hours per day times seven days per week) would require the station to operate with all chargers in use and little customer idling for 28 afternoon hours per week. If there is a large amount of stochastic variation in demand, then this may not be possible.<sup>22</sup> I therefore set  $\bar{X}$  as 70% of maximum capacity over a period to account for variation in demand. I discuss the implications of this choice when I present results in [Section 6.3](#).

An alternative to recovering fixed costs with linear prices is to use two-part tariff pricing. Electrify America, EVgo, and Tesla all offer subscription plans and are at least somewhat vertically integrated – Tesla sells EVs in addition to charging, Electrify America is owned by Volkswagen, an EV manufacturer, and EVgo has partnered with GM, another EV manufacturer. Subscriptions and vertical integration both allow firms to use a two-part tariff pricing model to increase profit relative to linear prices. With two-part tariffs, monopolists lower price to increase consumer surplus and raise the subscription or car price to extract the increased consumer surplus.

Networks could keep prices at the at those that maximize social surplus and use subscription fees to recover the remaining fixed costs (or generate larger profits) ([Feldstein, 1972](#); [Littlechild, 1975](#)). No network requires a subscription to use its stations,<sup>23</sup> so networks may choose to recover fixed costs by raising per kWh prices. I do not have any data on subscription prevalence or revenues. I therefore compare the per kWh prices subscribers pay to the set of surplus maximizing prices to test for networks’ use of two-part tariffs or vertical integration.

The goal of the remainder of this section is to calculate how far prices are from the social surplus maximizing and constrained Ramsey prices. In order to calculate these, I must derive own- and cross-price elasticities of demand to discipline substitution patterns as

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<sup>22</sup>It seems likely there is significant stochastic variation. Tesla has switched from time-of-use pricing to demand-based pricing, which would not be necessary if there was no stochastic variation in demand.

<sup>23</sup>California bans the requirement of subscriptions, but has no regulation on differences between subscriber and non-subscriber prices. Stations could charge much higher per kWh prices to non-subscribers to sidestep the policy, but the largest difference between subscriber and non-subscriber prices is 40%, so it appears networks have not chosen to effectively require subscriptions.

prices change. I also need the distribution of consumers to determine market demand and consumer surplus. I proceed by estimating a model of heterogeneous consumer demand to recover these.

## 6.2 Demand Estimation

### 6.2.1 Model

EV drivers must regularly choose where and when to recharge their vehicles. Preferences for charging location are likely based on a variety of factors including price, proximity, nearby amenities, time of day, charging speed, and reliability. Importantly, those preferences are likely to be heterogeneous across consumers. Therefore, my model builds on the framework for estimating models of discrete choices across heterogeneous consumers developed by [Berry \(1994\)](#) and [Berry, Levinsohn and Pakes \(1995\)](#).<sup>24</sup>

In each period  $t$  in market  $m$ , consumers choose when and where to charge their electric vehicle.<sup>25</sup> Consumer  $i$ 's indirect utility from charging their vehicle at a certain station during a specific hour, jointly denoted as product  $j$ , is given by the equation:

$$u_{ijt} = x_{jt}\beta_{it} - \alpha_{it}p_{jt} + \xi_{jt} + \varepsilon_{ijt} \quad (8)$$

where  $\alpha_{it}$  and  $\beta_{it}$  reflect preferences over prices,  $p_{jt}$ , and a vector of other observable station characteristics,  $x_{jt}$ , respectively.  $\xi_{jt}$  captures the utility from station-hour attributes that are observed by consumers and the network but not the econometrician.<sup>26</sup> Indirect utility also includes individual preference shocks  $\varepsilon_{ijt}$ . I assume  $\varepsilon_{ijt}$  are i.i.d (across  $(i, j, t)$ ) draws from a type I extreme value distribution, and are observed by neither the econometrician nor network. This specification yields the mixed Logit model.

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<sup>24</sup>I make three assumptions to consider charging quantities in a discrete framework. First, I make the same assumption as [Nevo \(2001\)](#), that consumers do not gain utility from consuming from multiple chargers. Their decisions can be seen as multiple purchase instances that are independent of each other, which avoids the need to model consumption as a continuous variable. Second, I assume that the choice set does not impact the amount that consumers drive. The total amount of gasoline purchased is highly inelastic, which is in line with this assumption: if the price of driving decreases, consumers will still drive the same amount. Third, I assume that consumers have decided on an optimal amount to charge before choosing their charging station. This is similar to [Hendel \(1999\)](#), who models firms buying multiple units of several brands of computers as making the optimal choice on many tasks, which in aggregate yield the observed purchase.

<sup>25</sup>No station hour can be in two markets, so I do not include  $m$  in notation because it is fully determined by station hour  $j$ .

<sup>26</sup>For example, suppose a charging station is located in the parking lot of a grocery store and the grocery store goes out of business. The station may anticipate reduced charging demand and respond by lowering prices. The econometrician would underestimate the price elasticity of demand because they would assume that the change in quantity was purely a function of the price change (if no other characteristics changed).

The mixed Logit allows preferences to vary across consumers and time. For example, individual  $i$ 's preferences over price and observable characteristic  $k$  are given by the following two equations:

$$\beta_{it}^{(k)} = \beta_0^{(k)} + \beta_\nu^{(k)} x_{jt}^{(k)} \nu_{ik} \quad (9)$$

$$\alpha_{it} = \alpha_0 + \alpha_\nu \nu_{ip} \quad (10)$$

where  $\beta_0$  and  $\alpha_0$  measure average utility across all consumers and the terms  $\beta_\nu^{(k)} x_{jt}^{(k)} \nu_{ik}$  and  $\alpha_\nu p_{jt} \nu_{ip}$  capture deviations in preferences from the population average. It is convenient to rewrite the utility as a function of average and individual preferences:

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt} \quad (11)$$

where  $\delta_{jt}$  is the mean utility consumers receive from choosing to charge at station-hour  $j$  during period  $t$ , and  $\mu_{ijt}$  is the individual (mean zero) deviation from that mean utility. I decompose the mean utility term into both observed and unobserved components:

$$\delta_{jt} = x_{jt} \beta_0 - \alpha_0 p_{jt} + \xi_{jt} \quad (12)$$

where  $\beta_0$  and  $\alpha_0$  reflect average utility from  $x_{jt}$ , a vector of  $K$  observable characteristics, and  $p_{jt}$ , the price per kWh of charging, respectively.  $\xi_{jt}$  captures the impact of unobserved station-hour attributes on utility. The individual deviation can be expressed as:

$$\mu_{ijt} = \sum_K [\beta_\nu^{(k)} x_{jt}^{(k)} \nu_{ik}] - \alpha_\nu p_{jt} \nu_{ip} \quad (13)$$

where  $K$  is the vector of observable characteristics. For simplicity, I will instead write  $\mu_{ijt} = (x_{jt}, p_{jt}) \cdot \Sigma \nu_{it}$  where  $\Sigma$  is a  $(K+1) \times (K+1)$  diagonal matrix with the diagonal  $(\alpha_\nu, \beta_\nu^1, \dots, \beta_\nu^K)$  and  $\nu_{it}$  is the vector of individual preference shocks.

Consumer  $i$  chooses to charge at station-hour  $j$  in period  $t$  if  $u_{ijt} \geq u_{ikt}$  for all  $k \neq j$ ,  $j, k \in m$ . One possible  $k$  is the outside option, charging at a slower charger or station not in my data, which gives utility  $u_{i0t} = \varepsilon_{i0t}$ . That is, consumers choose the station-hour that gives them the highest utility of all station-hours in market  $m$  during period  $t$ , unless this is less

than the utility from the outside option.<sup>27</sup> Integrating over all consumer choices using the distribution of consumer preferences,  $F(\nu_{it})$ , yields the market share for each station-hour:

$$s_{jt} = \sigma_j(\boldsymbol{\delta}_t, x_t, p_t; \Sigma) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^J \exp(\delta_{kt} + \mu_{ikt})} dF(\nu_{it}) \quad (14)$$

where  $\boldsymbol{\delta}_t = (\delta_{1t}, \dots, \delta_{Jt})$ ,  $x_t = (x_{1t}, \dots, x_{Jt})$ , and  $p_t = (p_{1t}, \dots, p_{Jt})$ .

### 6.2.2 Estimation

The goal of estimation is to recover the vector of parameters  $\theta = (\alpha_0, \beta_0, \Sigma)$ . One could theoretically estimate the parameters by minimizing the distance between the observed and predicted market shares. However, such an approach does not account for the correlation between prices and unobserved product characteristics,  $\xi_{jt}$ . To handle potential correlation, I implement the approach used by [Berry, Levinsohn and Pakes \(1995\)](#).

The first step is to recover mean utility,  $\delta_{jt}$ , for each product and time. The presence of  $\xi_{jt}$  means I cannot estimate  $\beta_0$  and  $\alpha_0$  directly and instead calculate mean utility. [Berry \(1994\)](#) and [Berry, Levinsohn and Pakes \(1995\)](#) show that one can fix the elements of  $\Sigma$  at predetermined values and invert the market shares to yield  $\hat{\delta}_{jt} = \hat{\sigma}_{jt}^{-1}(\hat{s}_{1t}, \dots, \hat{s}_{Jt}; \Sigma)$ .

The second step is to regress  $\hat{\delta}_{jt}$  on  $x_{jt}$  and  $p_{jt}$  to estimate  $\alpha_0, \beta_0$  using the equation  $\hat{\delta}_{jt} = x_{jt}\beta_0 - \alpha_0 p_{jt} + \xi_{jt}$ . However,  $\xi_{jt}$  are observed by consumers and the network but not the econometrician.  $\xi_{jt}$  likely impact prices and may be correlated with the observed variables  $x_{jt}$  (which are assumed to be exogenous), which would bias estimates of  $\alpha_0, \beta_0$ . The solution is to use a vector of exogenous variables,  $\mathbf{Z}_t$ , to build a vector of instruments,  $z_{jt}$ , for  $p_{jt}$ . One can then estimate an instrumental variables regression to recover  $\hat{\alpha}_0, \hat{\beta}_0$  (I explain my instruments in [Section 6.2.3](#)).

The estimation procedure explained so far relies on holding  $\Sigma$  fixed while one estimates  $\beta_0$  and  $\alpha_0$ . The identifying assumption for both the instrumental variables regression in the second step and estimation of  $\Sigma$  is  $E[\xi_{jt}|\mathbf{Z}_t] = 0$ , which implies  $E[z_{jt}\xi_{jt}] = 0$  for any  $z_{jt}$  constructed from  $\mathbf{Z}_t$ . The empirical analogue for the  $n^{th}$  moment is  $m^n(\theta) = \hat{E}[z_{jt}^n \xi_{jt}]$ . One can stack all the moment conditions to form  $\mathbf{m}(\theta)$  and create the generalized method of moments (GMM) estimator:

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<sup>27</sup>The majority of outside option charging is home charging, but the outside option also includes workplace charging and some charging at fast charging stations I do not observe.

$$\hat{\theta} = \arg \min_{\theta} \mathbf{m}(\theta)'W\mathbf{m}(\theta) \quad (15)$$

where  $W$  is a positive definite weighting matrix with dimension  $N \times N$ . [Conlon and Gortmaker \(2020\)](#) calculate a GMM objective  $q(\Sigma) = \mathbf{m}(\theta)'W\mathbf{m}(\theta)$  (note that  $q(\Sigma)$  does not include  $\alpha_0$  and  $\beta_0$ , which concentrated out from  $\theta$  because, given each  $\Sigma$ , they have a closed-form solution). They continue to try new  $\Sigma$  until the gradient of  $q$  with respect to  $\Sigma$  is approximately zero and one can conclude that  $\mathbf{m}(\theta)'W\mathbf{m}(\theta)$  has been minimized. At each guess of  $\Sigma$ , they repeat the first and second steps and evaluate  $q(\Sigma)$ .

Any positive definite  $W$  gives a consistent estimate of  $\theta$ , but only one  $W$  provides efficient estimates. The efficient weighting matrix is the inverse of the asymptotic covariance matrix of the population moments. Therefore, in order to get the efficient estimates, one must solve the problem twice. The first solution yields a consistent estimator that is used during the second iteration to recover the efficient estimator.

I conduct the estimation process using the *PyBLP* Python package from [Conlon and Gortmaker \(2020\)](#), which is based on the approach developed by [Berry, Levinsohn and Pakes \(1995\)](#). I estimate the mixed Logit model with markets defined at the week by CBSA level.<sup>28</sup> CBSAs are spatially distinct and there are very few stations that have a nearby station not in their CBSA, so it is unlikely that consumers are substituting across CBSAs. I choose to divide use into weeks to match my price data and create more moments to match in the estimation process. Because prices change infrequently and with no regularity, I argue it is unlikely that consumers are considering cross-week substitution.

I include different vectors of station characteristics for  $\theta_{\alpha_0, \beta_0}$  and  $\Sigma$ . I estimate homogeneous preferences for weekly price, reliability (station health index score over the prior four weeks), whether the station has a low, medium, or high health index score during the current week,<sup>29</sup> and station-hour fixed effects. The vector of characteristics estimated with heterogeneous preferences includes time of day (morning: 6-9 AM, late morning: 9 AM - 12 PM,

<sup>28</sup>I drop CBSAs without at least ten stations, which leaves seven CBSAs and 56 weeks totaling 392 markets.

<sup>29</sup>I use the categorical variable in an attempt to control for the share of chargers that were operable during the week rather than the share of charging attempts that were successful. Both are included in the calculation of the health index, and the effect of a broken charger on the index should be larger than the effect of a failed charging attempt. Drivers can learn of broken chargers from mobile applications and choose to avoid a station whereas they will suffer failures to charge at the station and may choose to use another charger at the station. I observe clumping just below health index scores of 50 and 75 in the data, which I believe is indicative that there are a large number of cases where one or two of four chargers are broken for the entire week and there is some variation in the number of failed charges that pushes the scores just below the corresponding share of chargers that are functional (for a station with three functioning chargers and one broken charger, this would be 75% which I expect maps to a health index score of 75).

afternoon: 12-4 PM, early evening: 4-7 PM, evening: 7-11 PM, and overnight: 11 PM - 6 AM) and charging network.

### 6.2.3 Instruments

I build a vector of instruments that incorporates both cost and demand shifters. Price is endogenous to demand, so higher priced products likely also have higher market shares. Regressing market shares on prices without an instrument would suggest that demand is increasing in price, even though we expect that it is likely decreasing in the price of a specific good. Instruments allow the econometrician to separate the endogenous and exogenous portions of price.

The goal of cost instruments is to capture the effect of marginal cost – rather than demand – on prices. I use the marginal cost of electricity that each station faces as an instrumental variable. This fully captures changes in marginal operating costs. There is variation in the marginal cost of electricity across hours and weeks, but there is very little within-week-hour across-station variation.

I rely on demand shifters to capture variation in exogenous price pressure across stations. My first set of demand instruments is a set of differentiation instruments, a geographical analog to those proposed by [Gandhi and Houde \(2020\)](#). I instrument for the number of own-network and competitor stations within 5-, 10-, 15-, and 20-minute rings of the station. These instruments are intended to account for the amount of differentiation and therefore price setting power a station has in the market. The final set of demand instruments is the price of residential charging at different times of the day to capture changes in relative preference for the outside option.

### 6.2.4 Results

I present the estimated parameters of the model of consumer demand in [Table 3](#). The aggregate preferences  $(\alpha, \beta)$  are calculated as the average preference across individuals. The average preference for reliability is positive and significant while the coefficients for stations with low or medium health scores are negative and significant. These results show that consumers value station functionality, which matches what one would expect. The average preference for price is negative and significant, suggesting general preferences for cheaper charging. These preferences are relatively small though. The implied willingness to pay for a 10 point increase in reliability (measured 0 to 100) is 2.6 cents and 16.8 cents for increasing the current week health index from below 50 to above 75.

Estimates of the nonlinear parameters  $(\sigma)$  are estimates of the standard deviation of pref-

Table 3: Estimated Demand System Parameters

| <i>Linear Parameters</i> ( $\alpha, \beta$ ) | Estimate | (SE)      |
|----------------------------------------------|----------|-----------|
| Price                                        | -2.655** | (0.648)   |
| Reliability                                  | 0.007**  | (0.001)   |
| Health Index < 50                            | -0.447** | (0.041)   |
| Health Index 50-75                           | -0.172** | (0.016)   |
| <i>Nonlinear Parameters</i> ( $\sigma$ )     |          |           |
| Morning                                      | -0.117   | (56.307)  |
| Late Morning                                 | 0.318    | (129.797) |
| Afternoon                                    | -0.930   | (66.499)  |
| Early Evening                                | 0.451    | (21.700)  |
| Evening                                      | -0.175   | (36.180)  |
| Overnight                                    | 7.671    | (4.323)   |
| Electrify America                            | 0.338    | (10.612)  |
| Tesla                                        | 0.041    | (8.970)   |

Notes: \*/ \*\* Denote significance at  $p < 0.05/0.01$  level.

erences across individuals. The estimates for time of day all have large standard deviations, which means I cannot rule out a large amount of heterogeneity in preferences for charging at different times of day. The results for network preferences are similarly inconclusive. One would expect that all or most of the  $\sigma$  parameters should be large. The results in [Section 5.2](#) suggest strong network preferences and we would expect consumers to potentially have rigid preferences for time of day as well. One possibility is that the lack of microdata or demographic data for fast charging consumers meaningfully reduces the precision of estimation. Another is that my data are noisy and also cannot capture the true demand at an hour where consumption is capacity constrained.<sup>30</sup>

I aggregate individual preferences to calculate elasticities for each individual and then average elasticities across individual. The model suggests that consumers have relatively homogenous price preferences, and are somewhat price sensitive. I present average own- and cross-price elasticities in [Table 4](#). Average own-price elasticities vary across networks and

<sup>30</sup>The capacity constraint both censors and influences demand. I cannot observe individuals who would have charged if there had been more chargers, and some individuals may see that a station is busy and choose to go somewhere else. [Conlon and Mortimer \(2013\)](#) develops a method for handling these types of cases, but without data on when chargers were occupied, I cannot implement it. It is not clear which direction this unobserved demand will bias my results. The inclusion of station-hour fixed effects means that one might expect that a given station-hour is more likely to be capacity constrained when prices are lower, leading to estimating consumers are less elastic than they truly are. On the other hand, [Figure 9](#) suggests networks may use prices as a rationing method meaning that demand would be underestimated at higher prices, leading to estimating that consumers are more elastic than they truly are.

Table 4: Average Own- and Cross-Elasticities

| Network           | Time          | Own     | Within<br>Station-Time | Within<br>Network | Within<br>Time |
|-------------------|---------------|---------|------------------------|-------------------|----------------|
| EVgo              | Morning       | -1.5268 | 0.0006                 | 0.0007            | 0.0013         |
| EVgo              | Late Morning  | -1.5922 | 0.0012                 | 0.0007            | 0.0029         |
| EVgo              | Afternoon     | -1.7087 | 0.0019                 | 0.0007            | 0.0053         |
| EVgo              | Early Evening | -1.7800 | 0.0014                 | 0.0007            | 0.0039         |
| EVgo              | Evening       | -1.6618 | 0.0008                 | 0.0007            | 0.0020         |
| EVgo              | Overnight     | -1.3681 | 0.0042                 | 0.0013            | 0.0080         |
| Electrify America | Morning       | -1.6080 | 0.0015                 | 0.0019            | 0.0013         |
| Electrify America | Late Morning  | -1.6062 | 0.0033                 | 0.0019            | 0.0030         |
| Electrify America | Afternoon     | -1.6038 | 0.0057                 | 0.0021            | 0.0056         |
| Electrify America | Early Evening | -1.6055 | 0.0040                 | 0.0019            | 0.0040         |
| Electrify America | Evening       | -1.6074 | 0.0021                 | 0.0019            | 0.0021         |
| Electrify America | Overnight     | -1.6014 | 0.0081                 | 0.0028            | 0.0084         |
| Tesla             | Morning       | -1.0553 | 0.0015                 | 0.0023            | 0.0013         |
| Tesla             | Late Morning  | -1.1692 | 0.0036                 | 0.0023            | 0.0029         |
| Tesla             | Afternoon     | -1.1815 | 0.0070                 | 0.0025            | 0.0055         |
| Tesla             | Early Evening | -1.1838 | 0.0050                 | 0.0023            | 0.0039         |
| Tesla             | Evening       | -1.1439 | 0.0026                 | 0.0023            | 0.0020         |
| Tesla             | Overnight     | -0.9887 | 0.0100                 | 0.0035            | 0.0083         |

time of day, with estimates ranging from -0.99 to -1.78 and an average elasticity of -1.33. The confidence intervals for these estimates are fairly large; the average elasticity has a 95% confidence interval of -0.66 to -1.87 (I provide confidence intervals for all own-price elasticity estimates in [Appendix Table 3](#)). These estimates are in line with the -1 to -1.45 [Bernard et al. \(2025\)](#) find in the United Kingdom. I estimate that cross-price elasticities are very low, suggesting little willingness to substitute across stations or hours. This results in a large amount of diversion to the outside good, which is likely charging at a home or slower public charger. Taken together, the own- and cross-price elasticities suggest consumers are somewhat price sensitive, but do not appear to search across stations for lower prices.

One additional takeaway from the demand estimation results is that elasticities vary across networks and time of day. The elasticity point estimates suggest Tesla stations face more inelastic demand EVgo and Electrify America stations, though the wide confidence intervals mean the differences are not significant. Estimates for within time of day substitution rates are generally highest overnight. This could reflect that charging during these times may disproportionately capture long distance and ride share driving, where the cost of delaying charging is greatest.

### 6.3 Comparison of Observed and Efficient Prices

I use the elasticities of demand and observed prices, costs, and quantities to solve for profit maximizing, social surplus maximizing, and constrained Ramsey prices. I assume linear demand and calculate profit maximizing prices as the set of prices that maximizes the profit portion of Equation (8) subject to the capacity constraint. These are static profit maximizing prices that would only maximize profit if competition is not dynamic. I argue this is a valid approach because the results in Section 5.2 suggest stations do not compete on prices or for the same customers and Figure 2 shows networks do not follow each other's price changes.<sup>31</sup> I calculate social surplus maximizing prices as the set of prices that maximizes Equation (8) subject to only the capacity constraint. The constrained Ramsey prices are the set of prices that maximizes Equation (8) subject to both the capacity constraint and restriction that profits exceed some level in order to recover fixed costs.

I start by comparing observed prices to both the set of prices that maximize profit and the set that maximize social surplus. In Figure 11, I plot the average prices and marginal cost by network and time of day, using the same divisions as in the demand estimation. I show both the average observed price that a network's subscriber would pay and that a driver without a subscription would pay. Across all networks and times of day, the average observed non-subscriber price is similar to the average profit maximizing price.<sup>32</sup>

The alignment of observed prices and estimated profit maximizing prices provides a measure of validation for the demand estimation results.<sup>33</sup> Setting non-subscriber prices at profit maximizing prices appears to be the optimal choice for networks. They may be able to increase profits by increasing consumer surplus to either raise subscription revenue or, if they are vertically integrated, car revenue. However, Tesla grants a free subscription to owners, so it has no incentive to charge below the profit maximizing price for non-subscribers. The other two networks do not give free lifetime subscriptions to car buyers (though they often give shorter duration free subscriptions). The alignment of their non-subscriber prices with the profit maximizing prices suggests that neither network is pursuing a vertical integration strategy. This makes sense – both networks are separate entities from car retailers and not truly vertically integrated.

The difference between marginal operating cost and the welfare maximizing prices reflects

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<sup>31</sup>The results in both Section 5.2 and Section 6.2.4 show consumers substitute very little across networks, so making the model dynamic would have little downward pressure on prices.

<sup>32</sup>The one exception to this is overnight at Tesla stations, when demand is more inelastic and prices are low. It is possible prices are lower to encourage consumers to switch from peak-demand hours to overnight hours or to avoid displeasure from consumers on roadtrips who need to charge overnight. Given that few drivers charge overnight, there is relatively little revenue to be made from raising prices during those hours.

<sup>33</sup>My demand estimation model does not have a supply side, so there is no assumption that the prices are profit maximizing.

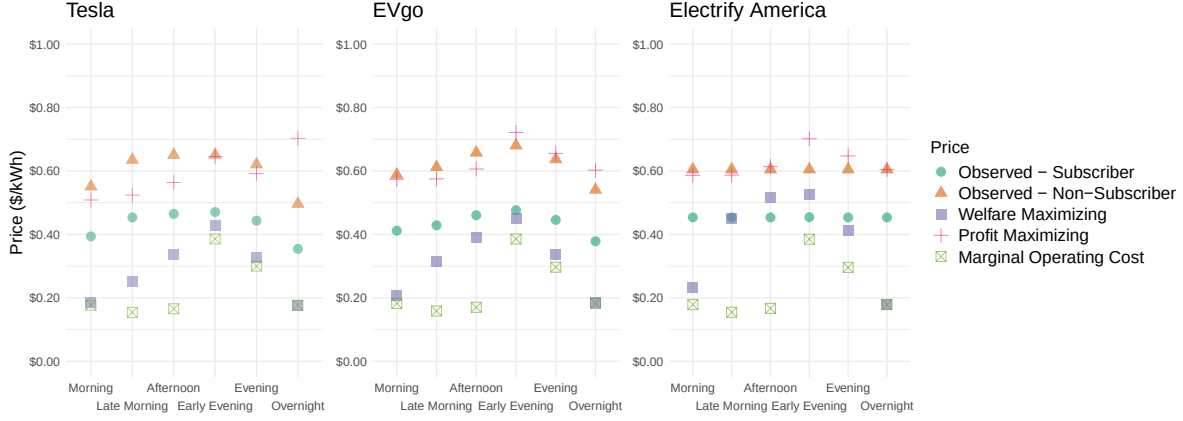


Figure 11: Comparison of Observed and Profit and Social Surplus Maximizing Prices

capacity constraints.<sup>34</sup> If a station is not constrained at a given time, then the social surplus maximizing price is equal to the marginal operating cost. When the station faces capacity constraints, the welfare maximizing price is greater than the marginal cost to allocate charging to consumers with the greatest willingness to pay. The more that capacity constraints bind during a time of day, the higher efficient prices will be at that time.

A second result evident in Figure 11 is that subscriber prices are generally above the welfare maximizing prices. Additionally, there is much more within-day variation in welfare maximizing price than in observed prices. This suggests that networks are not attempting to use a two-part tariff approach to maximize profit or recover fixed costs and instead focused on linear prices. An important comparison then is between observed prices and the prices that produce the greatest social surplus while still recovering a station’s fixed costs, the Ramsey prices. If prices are notably above the Ramsey prices, then there is room to increase social surplus by decreasing prices.

I do not observe the cost of building or maintaining stations, so I consider a range of fixed cost assumptions. Tesla claimed in 2025 that their average cost to build a station was about \$40,000-45,000 per charger (Miyatra, 2025), which I use as a low-end construction cost estimate because it is lower than academic estimates (Gamage, Tal and Jenn, 2023; Li, 2023). As the upper end, I use the 75<sup>th</sup> percentile of National Electric Vehicles Infrastructure (NEVI) site cost per charger, which is \$233,000 (McDonald, 2024). I use the 75<sup>th</sup> percentile because NEVI funding was limited to stations on designated alternative fuels corridors, which are likely to require more grid upgrade and construction costs than the stations in my data.

The AFDC estimates maintenance costs at about \$400 per charger month (AFDC, 2023) and I assume that stations pay monthly capacity charges for 200 kW per charger. Using a

<sup>34</sup>I show in Appendix Figure 19, that changing the capacity threshold from 70% to 60% or 80% of feasible maximum charging changes welfare maximizing prices by a few cents.

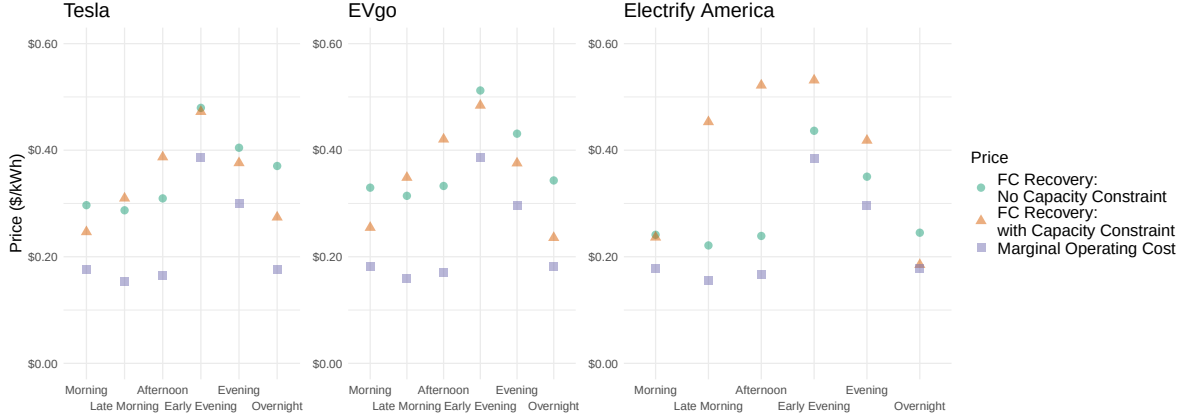


Figure 12: Comparison of Welfare Maximizing Prices, with and without Capacity Constraints

frequently assumed lifespan of 10 years and discount rate of 6%, I calculate the net-present value of lifetime costs as \$109,000 under the low-end estimate and \$302,000 under the upper-end estimate. For reference, [Li \(2023\)](#) and [Gamage, Tal and Jenn \(2023\)](#) estimate per charger costs at \$152,000-302,000 and \$122,000-440,000, respectively. My fixed cost estimates are slightly lower, which partially reflects that the stations in my sample are notably larger, 9.9 chargers per station, than those in [Li \(2023\)](#) (1.7 chargers per station) and [Gamage, Tal and Jenn \(2023\)](#) (1.5 chargers per station). I translate my bounds to weekly costs and set the amount of fixed-cost recovery for the low, medium, and high fixed cost scenarios at \$210, \$395, and \$580 per charger per week, respectively.

Capacity constraints put upward pressure on afternoon and evening prices. The Ramsey pricing literature shows that markups above marginal cost should be proportional to elasticities of demand. In [Figure 12](#), I plot the welfare maximizing linear prices that recover fixed costs under a medium fixed cost assumption. In the absence of capacity constraints, the markups would be fairly consistent throughout the day. However, incorporating capacity constraints pushes up efficient prices during congested times of day to ensure market clearing.

In [Figure 13](#), I plot observed and Ramsey prices under different fixed cost assumptions. I enforce the capacity constraint across all three fixed-cost recovery scenarios, so afternoon prices remain well above marginal operating cost for all scenarios. Subscriber prices appear to be in line with the Ramsey prices for the high fixed cost scenario. There appears to be some room to lower non-subscriber prices then, but not a large amount. It is worth noting that not all stations successfully recover fixed costs under these scenarios. For example, [Appendix Figure 20](#) shows that only half of EVgo stations would be able to recover high fixed costs. All Electrify America and the vast majority of Tesla stations recover fixed costs across all three scenarios. The difference between EVgo and the other two networks

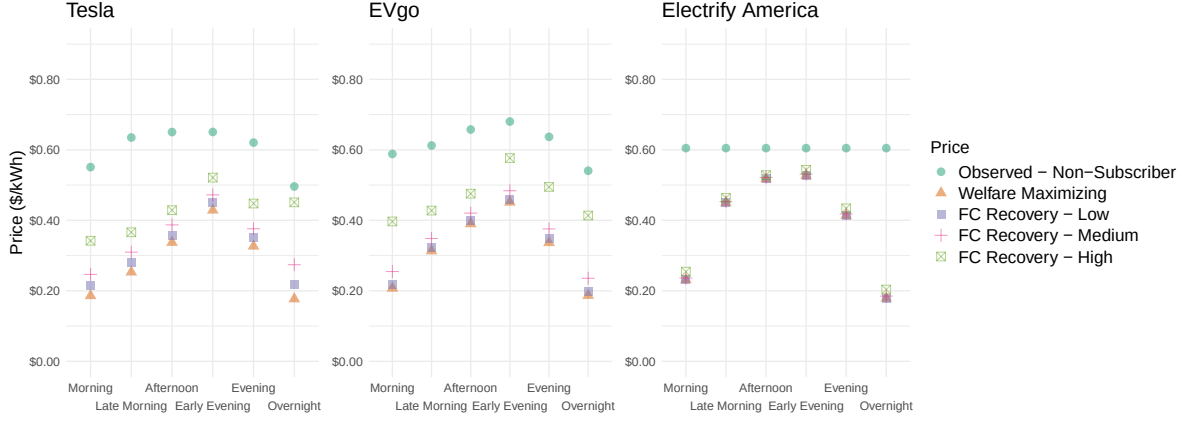


Figure 13: Comparison of Observed and Ramsey Prices

may reflect that their stations often use lower powered chargers (which are cheaper to build and require lower monthly capacity charges) and are more likely to receive state or federal subsidies.

I plug the Ramsey prices back into the demand estimation model to calculate the effects on surplus. I estimate that charging the Ramsey prices instead of observed prices would increase social surplus by 7-10% depending on the level of fixed costs. The increase to consumer surplus is much larger (31-72%), with slightly less than half coming from recovered deadweight loss. The remaining increase is a large transfer of surplus from producers, who lose 9 to 32% of observed producer surplus. This transfer would reduce networks' incentives to build stations. Policy aimed at lowering prices then could lead to reductions in charger supply.

No network appears to follow a two-part tariff or vertical integration strategy. Two-part tariffs are more efficient than linear prices for fixed-cost recovery and often notably reduce deadweight loss. Figure 13 shows that the optimal linear prices are almost equal to the efficient prices, and two-part tariffs provide a minimal social surplus improvement over Ramsey prices. This suggests that competition driving prices down could achieve nearly all of the social surplus benefits of networks using two-part tariffs pricing.

## 7 Conclusion

The EV fast charging market is a growing market. While many microeconomic models would suggest an increasing number of competitors would drive down prices, prices have stayed high as stations have entered. I show that this is because consumers and stations do not respond to the entry of new stations in competing networks. Under my preferred specification, I find stations do not lower prices in response to competitor entry and can

rule out price responses as large a share of margins as [Davis, McRae and Seira \(2025\)](#) and [Taylor and Muehlegger \(2025\)](#) find in at gas stations. Tesla stations do respond to nearby own-network entry by lowering prices. I show that this may be because new stations relieve capacity constraints at incumbent stations, so the market clearing price decreases.

The consumer response results may explain the mechanisms that lead to these price results. I estimate that quantities decrease at incumbents following own-network entry as the entrant likely pulls some consumers away from the incumbent station. However, entry has no effect on charging activity at incumbents. This may suggest a high degree of market segmentation as drivers appear to only respond to new stations from the network whose stations they already use. I show that this segmentation is not due to differences in charging standards across stations. Segmentation could be fostered due to subscription plans, the requirement to open a network’s mobile application to find prices (and often even charger availability), or reflect that some cars came with some amount of free charging at specific networks during the early 2020s, which could lead to inelastic demand.

Given the lack of price competition and prevailing high markups above marginal operating costs, I investigate how far prices are from surplus maximizing prices. I estimate a model of demand to recover own- and cross-price elasticities and find own-price elasticities ranging from -0.99 to -1.78 across networks and times of day. My estimates are consistent with the -1 to -1.45 [Bernard et al. \(2025\)](#) estimate. I embed the elasticities into a welfare maximization problem to determine the set of prices that maximizes social surplus while recovering fixed costs.

I show that the surplus maximizing prices are not set at marginal operating cost, which would lead to inefficient allocation when more drivers want to consume more than a station can charge. Instead, efficient prices need to account for capacity constraints, which appear to be impacting networks’ pricing and station construction choices. Market clearing prices that eliminate deadweight include 48% (quantity-weighted) average markups above marginal operating costs.

Observed prices are very similar to the set of linear prices that would maximize profits. There is no evidence that networks are using a two-part tariff system or vertical integration to maximize profits. The choice to use linear prices creates deadweight loss, but I show that at low fixed costs, the deadweight loss is minimal because the most efficient linear prices and two-part tariff prices nearly coincide. As fixed costs increase, the deadweight loss from use of linear prices increases. I estimate that switching to Ramsey prices would increase social surplus by 7-10% depending on the level of fixed costs.

The paper’s title asks what is driving high EV fast charging prices? The quantity-weighted average markup above marginal operating cost is 109%. A 48% markup is needed

to eliminate deadweight loss from misallocation. Beyond that, recovering fixed costs with linear prices requires a 53-98% markup depending on the fixed cost assumptions. Capacity constraints and fixed-cost recovery only explain about one third and one to two thirds of observed markups, respectively. The answer then appears to be that a lack of competition due to rigid consumer preferences is driving high markups.

The most common policy tool to address high prices is increasing competition. However, the entry analysis revealed that increased competition does not reduce prices (at least at the station level). Direct price regulation would also appear to be a poor choice: stations need to be able to raise prices when demand is high to ensure charging is allocated efficiently and encourage substitution to other stations or times of day. Focusing on the lack of substitution may provide a direction for policy makers worried about high prices. It's possible that the high search costs for finding stations and prices may deter substitution. Regulation that requires stations to make prices accessible through an API so that third-party applications can post prices for all stations could increase competition. Any policies that successfully lower prices could lead to decreased station entry, although reducing search costs could also reduce barriers to entry for smaller networks.

My results suggest that a large share of markups are efficient due to insufficient supply. Strong station and time preferences lead to relatively-low off-peak station usage while exhausting capacity during peak-demand periods. Policy focusing on fast charging may want to not only address the quantity of charging stations, but attempt to incentivize more utilization of charging stations during off-peak hours to reduce average fixed costs in the market and lower peak prices. Potential solutions include reducing search costs for consumers in the hope this increases cross-hour substitution and changing the structure of charging station electric tariffs.

Data on the fast charging market is sparse. In this paper, I am able to delve into prices in the fast charging market due to a large data collection effort. However, more data is needed on the costs that stations face and the decisions consumers make to understand the value of government investment in fast charging infrastructure and assess charging station policies. For example, consumer choice data would allow researchers to develop demand models that paint a more complete picture of how drivers substitute between charging locations. Given that low substitution is driving the need for more fast charging infrastructure due to capacity constraints at peak-demand hours, this research could be highly impactful. Hopefully researchers will be able to gain access to more fast charging data to answer questions about the value of stations and the most efficient tools for governments that want to facilitate a transition to electric vehicles.

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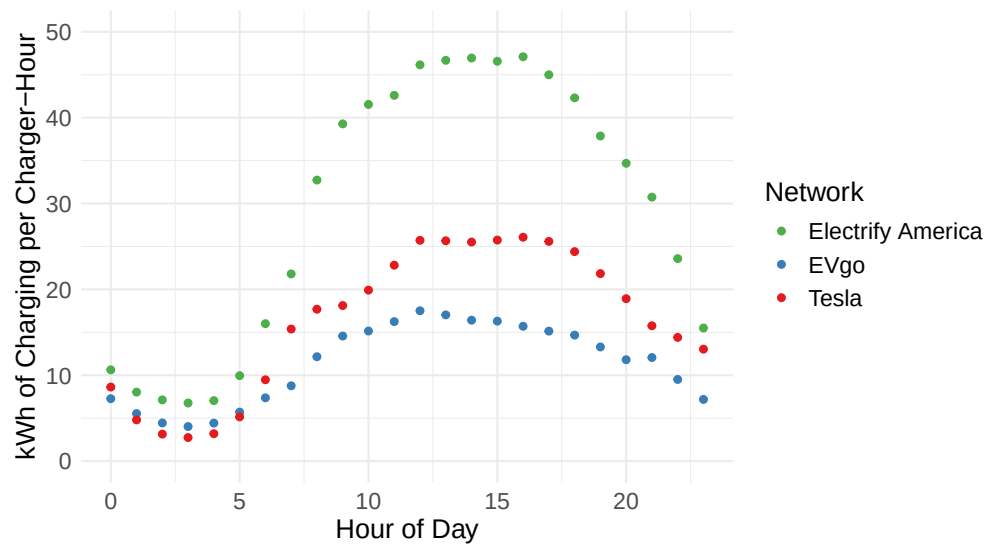
# A Appendix

Appendix Table 1: Charging Levels

| Charger Level    | Typical Power Output | Estimated BEV Charge Time | Estimated Electric Range per Hour of Charging | Typical Location            |
|------------------|----------------------|---------------------------|-----------------------------------------------|-----------------------------|
| Level 1          | 1 kW                 | 40-50 hours               | 2-5 miles                                     | Home                        |
| Level 2          | 7-19 kW              | 4-10 hours                | 10-20 miles                                   | Home, Workplace, and Public |
| DC Fast Charging | 50-350 kW            | 20 - 60 min.              | 180-240 miles                                 | Public                      |

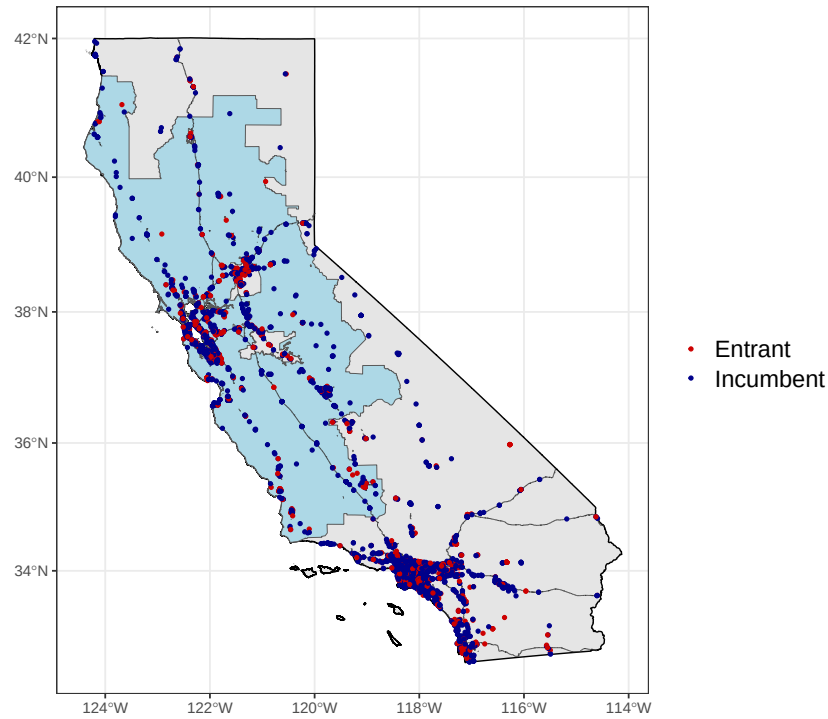
*Notes:* Source: U.S. Department of Transportation.

Charge time is calculated assuming the user charges a 60-kWh battery from 20-80% capacity. Most of the EV models popular in the U.S. have battery capacities around 60-80 kWh.



*Notes:* Data is for October 2025. Electricity utilization per charger is calculated at each station and averaged across all stations operated by each network in PG&E's service area.

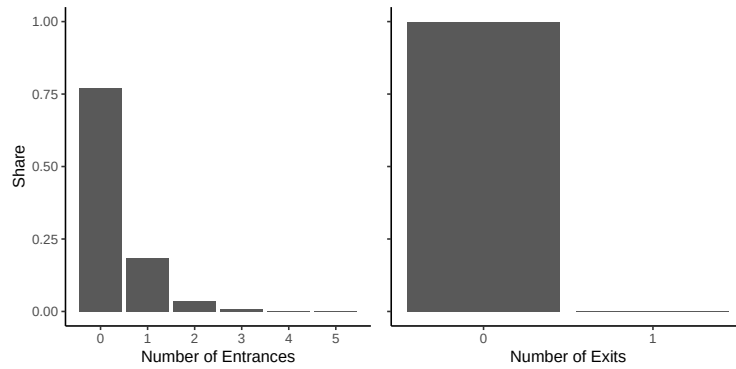
Appendix Figure 1: Average Hourly Electricity Use per Charger at DCFC Stations in PG&E Service Area



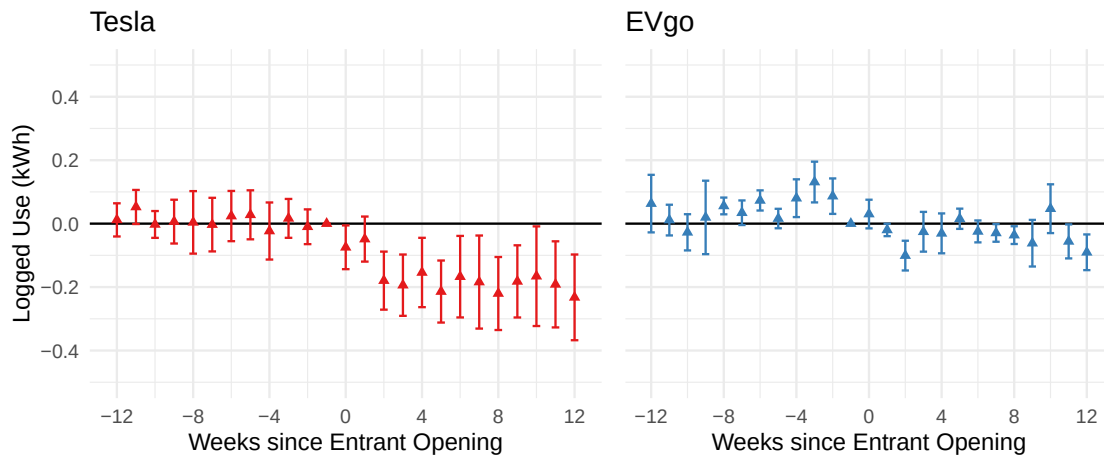
Appendix Figure 2: Map of Incumbent and Entrant Station Locations

Appendix Table 2: Entrant Characteristics

|                                          | Tesla | EVgo | Electrify America | Other |
|------------------------------------------|-------|------|-------------------|-------|
| Nearby (Total) Incumbent Characteristics |       |      |                   |       |
| Own stations                             | 0.2   | 0.4  | 0.1               | 0.2   |
| Competing stations, different standard   | 1.0   | 0.3  | 0.8               | 0.2   |
| Competing stations, same standard        | 0.0   | 0.7  | 0.7               | 1.3   |
| Own chargers                             | 3.0   | 1.3  | 0.5               | 0.4   |
| Competing chargers, different standard   | 3.4   | 4.4  | 16.4              | 2.6   |
| Competing chargers, same standard        | 0.0   | 3.1  | 1.5               | 5.7   |
| Entrant Characteristics                  |       |      |                   |       |
| Chargers                                 | 13.1  | 6.0  | 5.7               | 3.9   |
| First four weeks kWh/day                 | 4312  | 607  | 429               | 219   |
| First four weeks kWh/charger/day         | 333   | 97   | 83                | 65    |

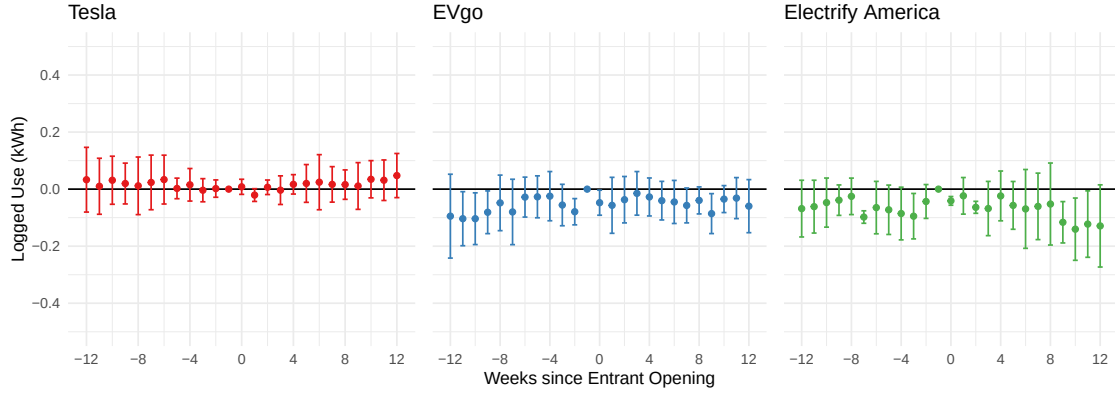


Appendix Figure 3: Frequency of Number of Entries and Exits Experienced by Incumbents



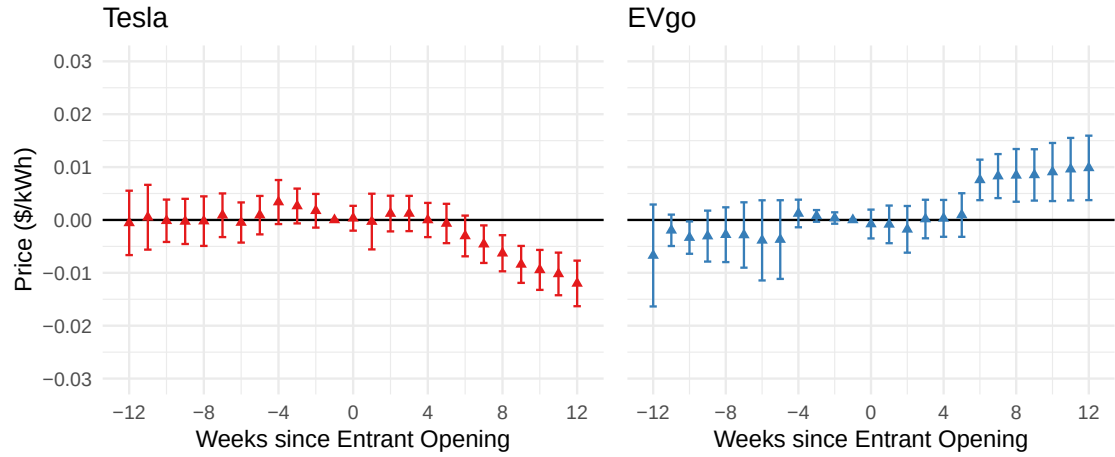
*Notes:* Data are total electricity consumption at each station and spans January 2022 - December 2025. Nearby entry is limited to stations within five minutes of an entrant. Electrify America does not open enough stations near other Electrify America stations to estimate an effect for the network.

Appendix Figure 4: Event Study Estimates of Own-Network Station Entry on Incumbent Hourly Electricity Consumption, by Network



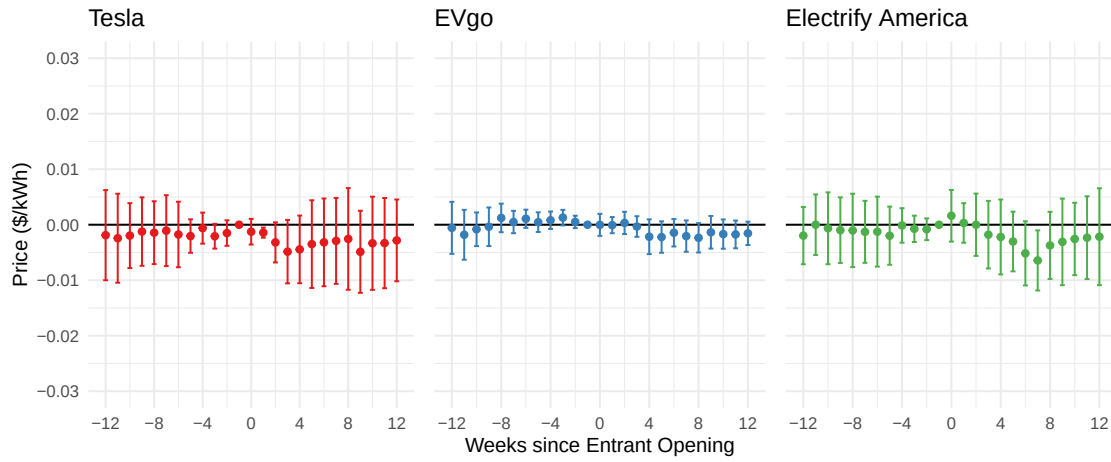
*Notes:* Data are total electricity consumption at each station and spans January 2022 - December 2025. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 5: Event Study Estimates of Competitor Station Entry on Incumbent Hourly Electricity Consumption, by Network



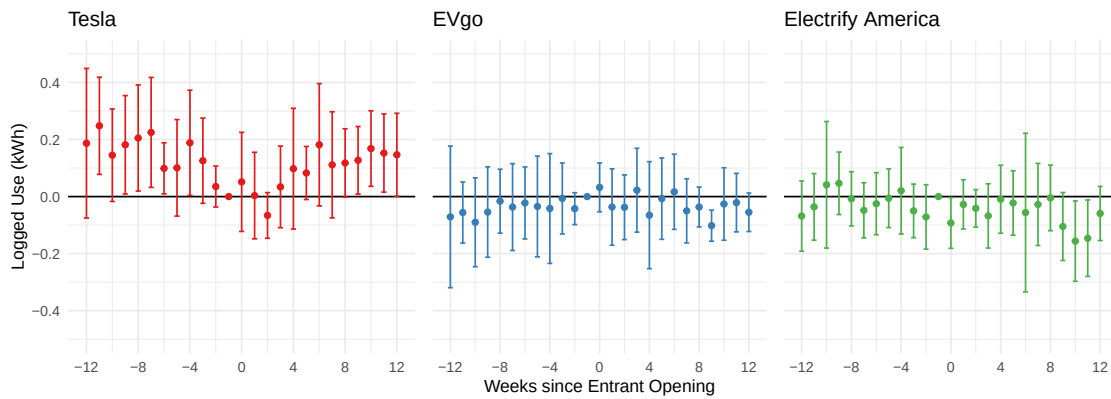
*Notes:* Data are station-week-hour prices and spans October 2024 - May 2026. Nearby entry is limited to stations within five minutes of an entrant. Electrify America does not open enough stations near other Electrify America stations to estimate an effect for the network.

Appendix Figure 6: Event Study Estimates of Own-Network Station Entry on Incumbent Hourly Prices, by Network



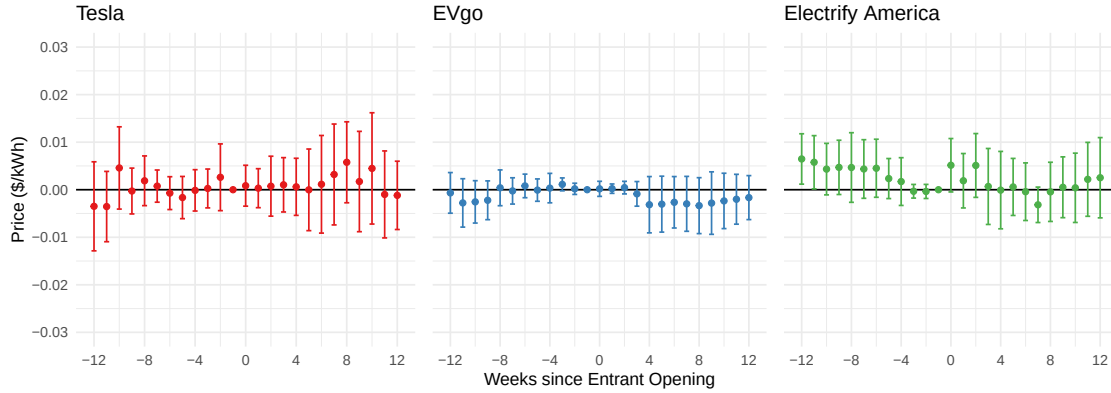
*Notes:* Data are station-week-hour prices and spans October 2024 - May 2026. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 7: Event Study Estimates of Competitor Station Entry on Incumbent Hourly Prices, by Network



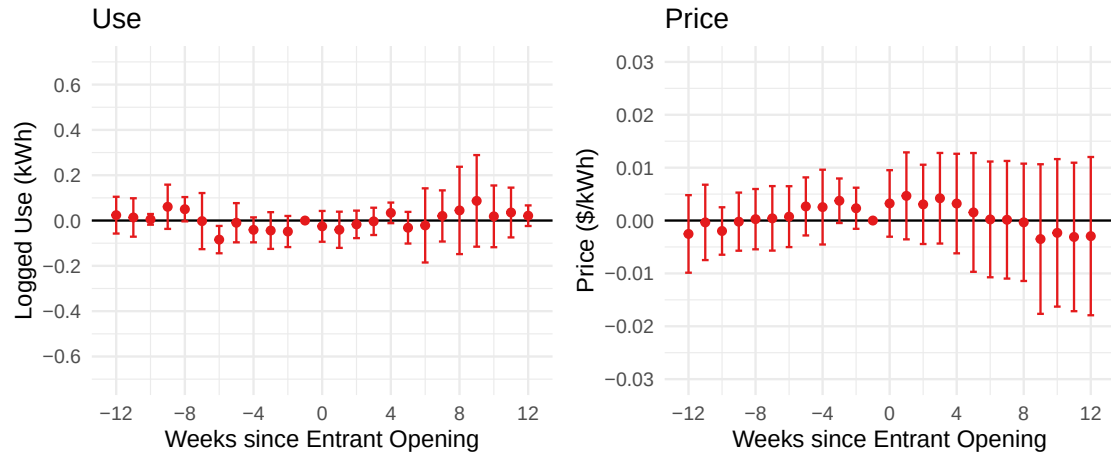
*Notes:* Data are total electricity consumption at each station and spans January 2022 - December 2025. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 8: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Five Minutes on Hourly Electricity Consumption, by Network



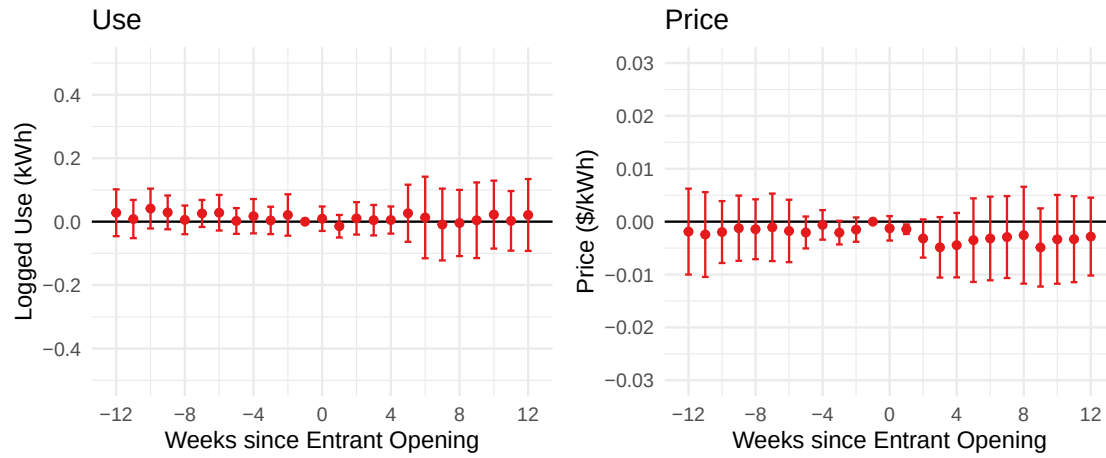
*Notes:* Data are station-week-hour prices and spans October 2024 - May 2026. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 9: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Five Minutes on Hourly Prices, by Network



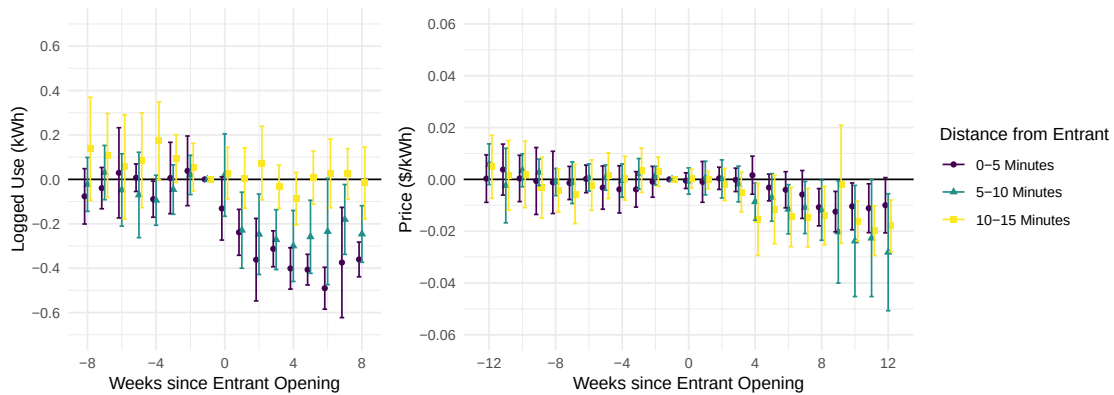
*Notes:* Data are hourly electricity consumption and prices at each station and span January - December 2025 and January 2025 - May 2026, respectively. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 10: Event Study Estimates of Charging Use and Price at Tesla Stations of Station Adding NACS Chargers



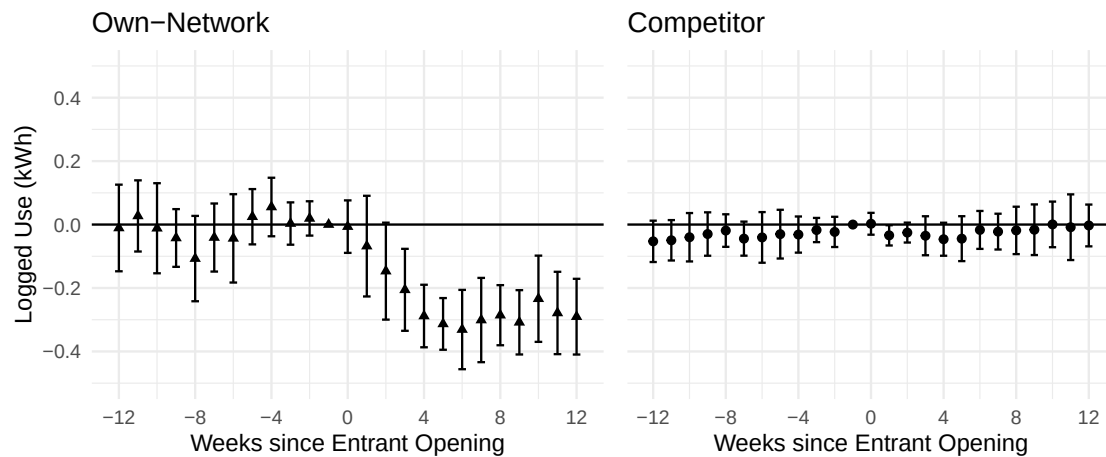
*Notes:* Data are hourly electricity consumption and prices at each station and span January - December 2025 and January 2025 - May 2026, respectively. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 11: Event Study Estimates of Charging Use and Price at Tesla Stations of Entrant in 2025 or Later



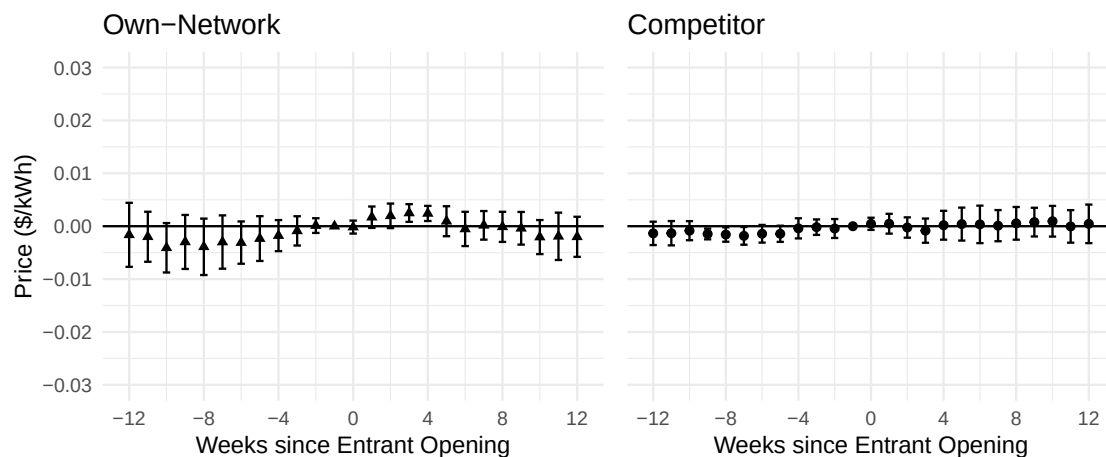
*Notes:* Data are hourly electricity consumption and prices at each station and span January - December 2025 and January 2025 - May 2026, respectively. Nearby entry is limited to stations within five minutes of an entrant.

Appendix Figure 12: Event Study Estimates of Tesla Own Station Entry within Five Minutes on Incumbent Hourly Electricity Consumption and Price by Distance



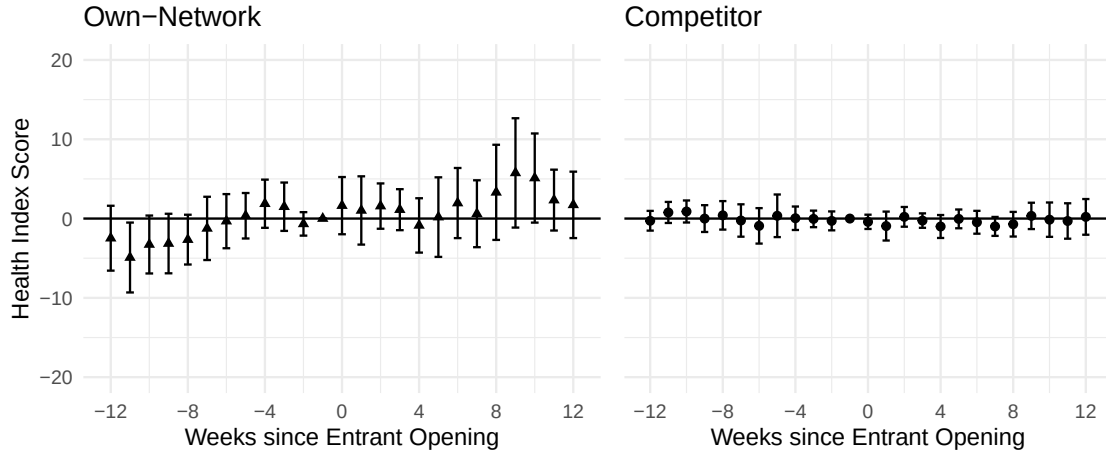
*Notes:* Data is for total electricity consumption at each station and spans January 2022 - December 2025.

Appendix Figure 13: Event Study Estimates of Own-Network Station Entry within Ten Minutes on Incumbent Hourly Electricity Consumption



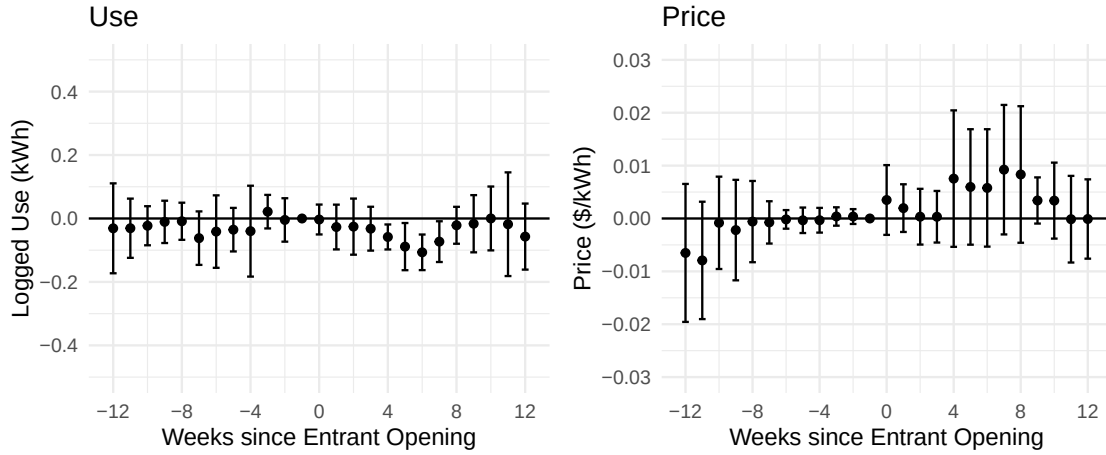
*Notes:* Data is for station-week-hour prices and spans October 2024 - May 2026.

Appendix Figure 14: Event Study Estimates of Own-Network Station Entry within Ten Minutes on Incumbent Price



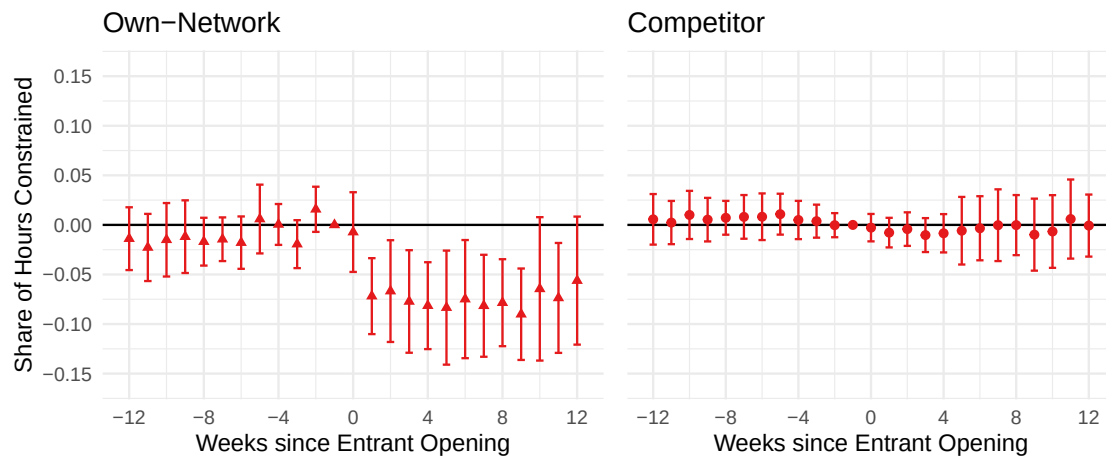
*Notes:* Data is for station-week health index score and spans October 2024 - May 2026.

Appendix Figure 15: Event Study Estimates of Station Entry within Ten Minutes on Station Quality



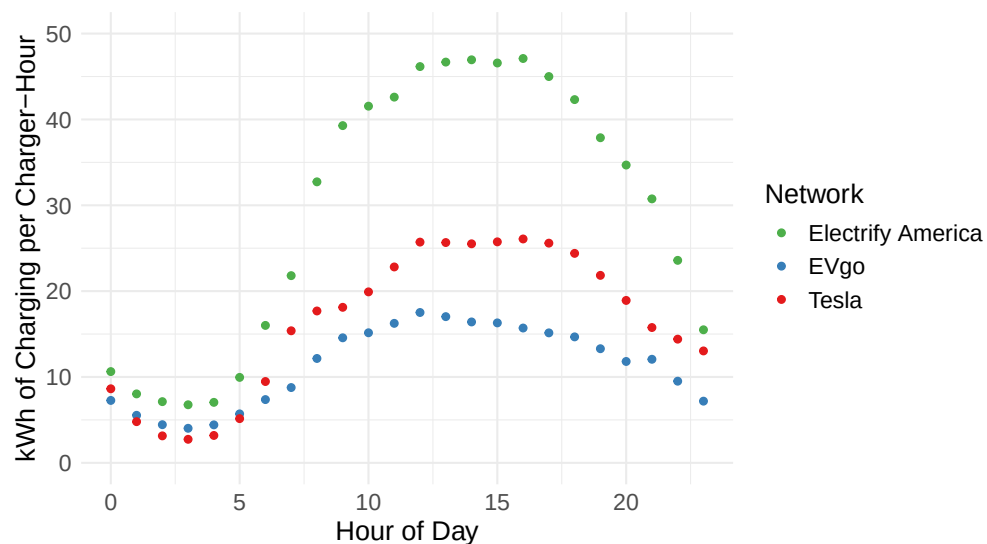
*Notes:* Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Appendix Figure 16: Event Study Estimates of Entry of Competing Station with Same Charging Standard within Ten Minutes



*Notes:* Data is for total hourly electricity consumption at each station and spans January 2022 - December 2025.

Appendix Figure 17: Event Study Estimates of Station Entry within Ten Minutes of Tesla Station on Share of Hours near Charging Capacity

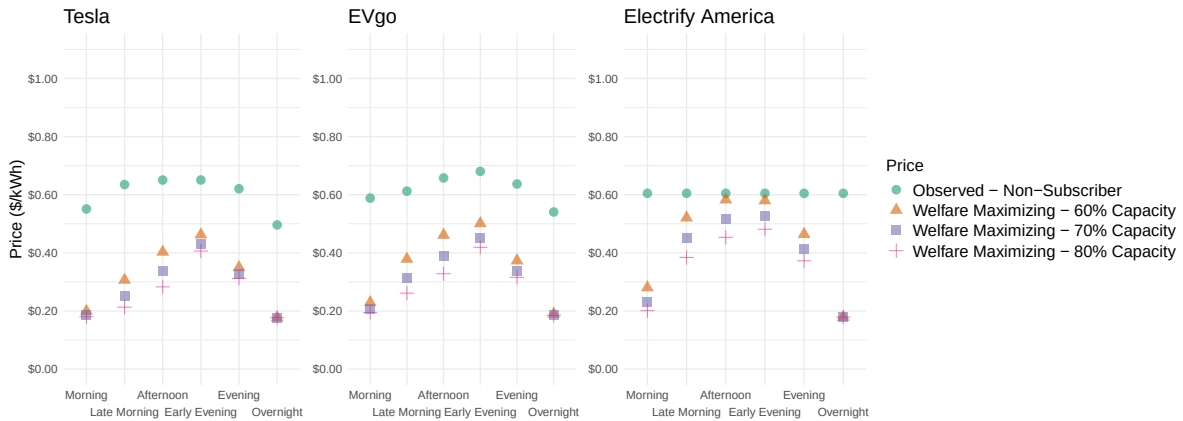


*Notes:* Data are for October 2025. Electricity utilization is the charger-weighted average across all stations operated by each network in PG&E's service area.

Appendix Figure 18: Average Hourly Electricity Use per Charger at DCFC Stations in PG&E Service Area

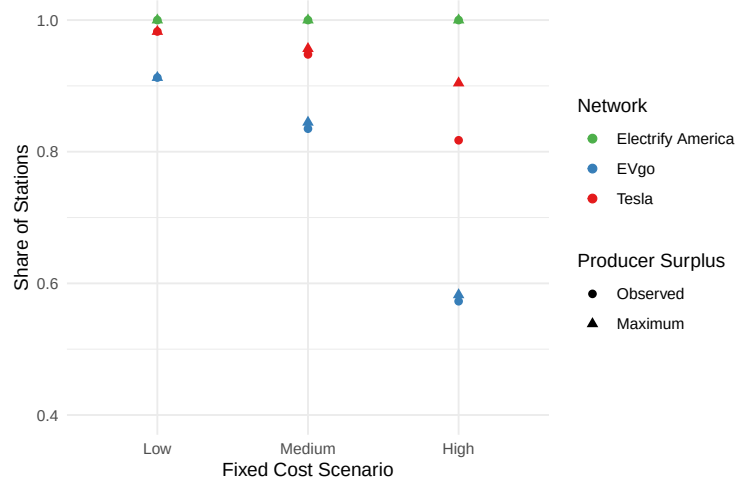
Appendix Table 3: Own-Price Elasticity Estimates and Confidence Intervals

| Network           | Time          | Central Estimate | Confidence Interval |
|-------------------|---------------|------------------|---------------------|
| All               | All           | -1.334           | (-0.66, -1.886)     |
| EVgo              | Morning       | -1.527           | (-0.722, -2.087)    |
| EVgo              | Late Morning  | -1.592           | (-0.749, -2.152)    |
| EVgo              | Afternoon     | -1.709           | (-0.83, -2.356)     |
| EVgo              | Early Evening | -1.780           | (-0.852, -2.421)    |
| EVgo              | Evening       | -1.662           | (-0.806, -2.3)      |
| EVgo              | Overnight     | -1.368           | (-0.7, -1.991)      |
| Electrify America | Morning       | -1.608           | (-0.748, -2.134)    |
| Electrify America | Late Morning  | -1.606           | (-0.75, -2.134)     |
| Electrify America | Afternoon     | -1.604           | (-0.767, -2.191)    |
| Electrify America | Early Evening | -1.605           | (-0.751, -2.138)    |
| Electrify America | Evening       | -1.607           | (-0.77, -2.186)     |
| Electrify America | Overnight     | -1.601           | (-0.801, -2.315)    |
| Tesla             | Morning       | -1.055           | (-0.523, -1.5)      |
| Tesla             | Late Morning  | -1.169           | (-0.58, -1.665)     |
| Tesla             | Afternoon     | -1.182           | (-0.594, -1.704)    |
| Tesla             | Early Evening | -1.184           | (-0.591, -1.693)    |
| Tesla             | Evening       | -1.144           | (-0.576, -1.644)    |
| Tesla             | Overnight     | -0.989           | (-0.507, -1.455)    |



*Notes:* Capacity percentage is how the the percentage of maximum feasible charging used to define capacity. Setting capacity at 60% of maximum feasible charging makes the capacity constraint bind at lower quantities than setting capacity at 70 or 80%.

Appendix Figure 19: Welfare Maximizing Prices under Alternative Capacity Constraints



Appendix Figure 20: Share of Stations Able to Recover Fixed Costs

## B Data Appendix

### B.1 Price Data

Each week, a team of research assistants and I collected hourly prices for all Tesla, EVgo, and Electrify America DCFC stations in California. Data comes from each company’s mobile application ([Appendix Figure 21](#) provides a visual of how prices look in the applications) and PlugShare, a third party aggregator of charging station data, which posts EVgo prices. Each of these sources lists prices for each hour during the given day, which allowed collectors to pull all prices for a station at once.

The list of stations comes from querying the Alternative Fueling Station Location API from the DOE’s Alternative Fuels Data Center (AFDC) each Saturday. Each data collector would check the hourly prices for a station from the list and enter all 24 prices into a spreadsheet for that week. Data collectors would repeat this process for all stations each week.

The collection period ran from Saturday to Friday each week. Each Saturday I would collect the list of stations from the AFDC, add them to a spreadsheet, and send out a list of collection assignments. Research assistants were told the deadline for data collection was Thursday, and were in charge of collecting prices for around 100 stations between Sunday and Thursday (there were usually four students collecting EVgo prices and six students collecting Tesla prices, and I collected all Electrify America prices). Friday acted as a buffer day to allow me to collect any missing data and spot check the collected data.

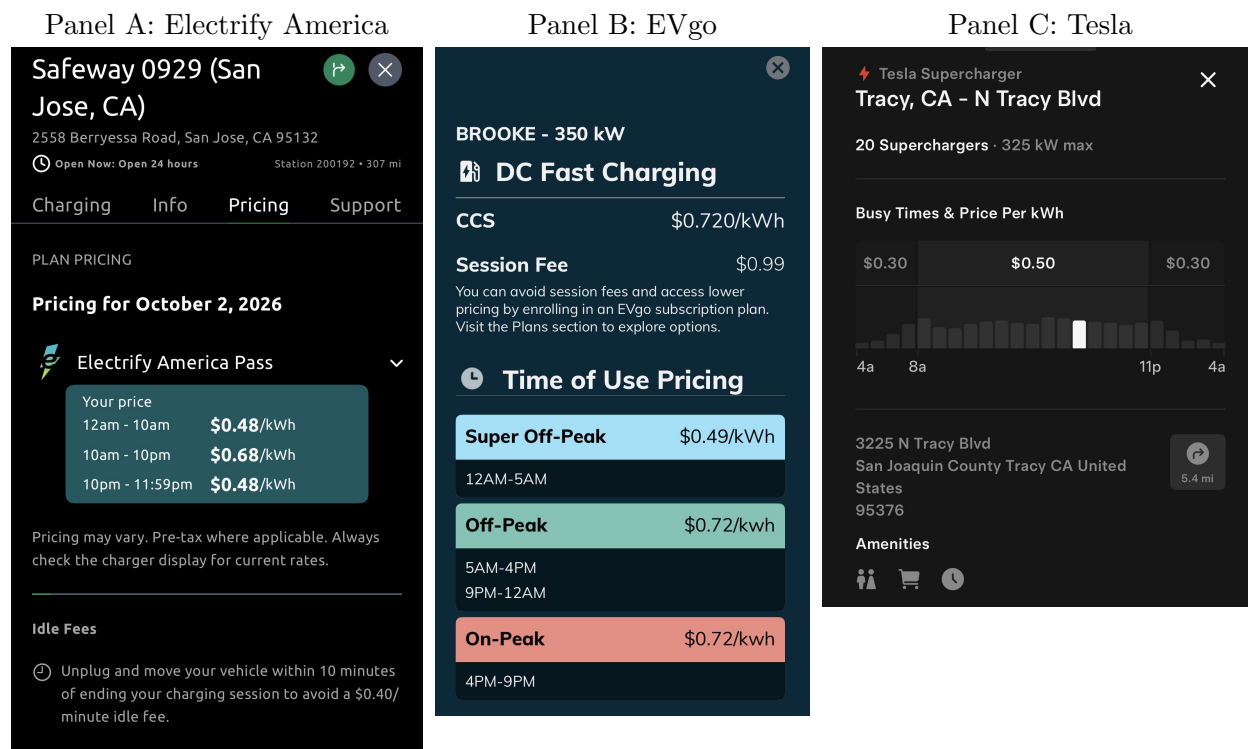
I piloted the data collection process from July through September of 2024 by collecting prices for Southern California stations. This helped create processes for checking data veracity. The data collection spreadsheet included a mechanism to compare against the prior week’s prices to make error identification easier. However, there are likely still errors in the data. I compared my data against data from Paren for September of 2025 and found a greater than 99% match rate across all station-hours during the month, which suggests errors were very infrequent.

I have good coverage of stations – the data include prices for 1407 out of 1414 Electrify America, EVgo, and Tesla stations listed in the AFDC data at any point between October 2024 and May 2026. I do not have data on the other seven stations because they did not list prices on their network’s application. This most likely means that the AFDC list includes some stations that are not operational and not that networks did not post prices for all stations. My final dataset cover 657 Tesla, 454 EVgo, and 296 Electrify America stations from October 2024 through May 2026, amounting to 2.6 million station-week-hour prices.<sup>35</sup>

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<sup>35</sup>We also collected prices for an additional 52 one-charger EVgo locations that I do not include in this

Prices could be collected any day between Saturday and Friday, and data collectors did not all choose to collect prices the same day. Therefore, it is important to be clear on interpretation of a week's prices as point in time prices collected during that week. My recorded weekly prices at a station will not capture price changes that occurred that week if the changes occurred after data was collected. Further, it is possible that during a week where a network changed many stations' prices, some data collectors recorded new prices while some recorded old prices.



Appendix Figure 21: Price Presentation in Mobile Applications

## B.2 Utility Data

Pacific Gas & Electric Company (PG&E) provided metered electricity use data for this project. They compiled all data for meters linked to the list of station addresses from the AFDC and all meters flagged in their data as providing public vehicle charging. Limiting to just stations in PG&E's territory, I am able to match 59% of stations from my price data (EVgo, Electrify America, and Tesla) and 43% of stations from the AFDC data (all stations).

analysis because I focus on locations with two or more chargers.

The low match rates reflect both that I cannot identify all stations in the PG&E data and that some stations may not be served by PG&E. It is possible that even stations PG&E serves will not be in the data they share with me because the station's meter is not at the address listed in the AFDC and the meter is not flagged as for a charging station. Because PG&E sent me data for all meters at the addresses of stations and many are located in large shopping developments, I have meters that are not used for stations. PG&E gives a 1-5 score for likelihood the meter is connected to a charging station. I only keep meters when it is clear that a station is the only customer connected to that meter. This means I am not able to identify many smaller stations that share meters with other businesses and some cases where multiple stations share a parking lot.

One additional caveat is that individual accounts may have multiple meters. Many stations have meters replaced at some point during the sample period, so I build a crosswalk of meters to stations. I aggregate meter data to the station level and use the AFDC station identifier to PG&E station identifier to merge price and use data.

I aggregate all data to the meter-hour level because some meters report hourly and some report every fifteen minutes. The electricity use is raw, and PG&E makes corrections to it before sending bills to customers. However, corrected billing volumes are only available aggregated to the billing period, so I use the raw volumes and acknowledge that they may introduce additional noise into the data.

To calculate hourly charging, I subtract out the baseline usage for each station. I calculate baseline hourly usage as the lowest use at a station. If a station has drivers charging for the entire sample period, I would overestimate the baseline, but this is highly unlikely as there is very little use overnight. If a station's lights are on the same meter as the chargers and are only on overnight, I will slightly underestimate the amount of overnight charging. The amount of electricity the lights use should be relatively low compared to the amount needed to charge a car. I also considered using a baseline calculated as the lowest usage during an hour of the day to avoid this problem. The results were similar between the two, and the latter option increased the probability that I set the baseline too high for station-hours that are consistently busy. It is much more likely that a station always has customers at 2 PM than that it always has customers at all hours.

The panel of station-week-hour charging contains 1.5 million observations for 468 stations over 209 weeks. 178 stations are Tesla stations, 128 are EVgo, 90 are Electrify America, and 72 belong to other networks. I am able to match all to price data, but can only include 115 Tesla, 103 EVgo, and 38 Electrify America stations in the demand estimation model. There are two reasons for the large drop in stations for Tesla and Electrify America. First, Electrify America installed batteries at many of its stations and I omit all stations with

batteries installed because batteries are charged in the late morning (when electricity is cheap) and discharged in the late afternoon (when it is expensive), which will bias estimates. Second, I limit the model to just CBSAs with at least one station from each network and at least 10 stations overall. Many Tesla stations are along highways in more rural areas and therefore are not included in the demand estimation.